



WOMEN'S SAFETY AUDIT OF TONDIARPET

FINAL REPORT

Prepared by Safetipin

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WOMEN'S SAFETY AUDIT
OF TONDIARPET

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TONDIARPET

WOMEN SAFETY AUDIT OF PUBLIC SPACES

Safety, accessibility and inclusion in public space

Public spaces are an integral part of our cities and play a pivotal role in enhancing liveability. Therefore, careful design and thoughtful management of public spaces can significantly impact liveability of urban areas. Well planned and designed public spaces in urban settings can improve social inclusion, civic participation and enhance overall safety for all groups in the city. To ensure that all groups can equally access and use public spaces and other public services in the city, it is essential to build a safe and inclusive environment with a focus on public transport.

Presence of safe public spaces and public transport in cities enable people to access various opportunities and enjoy their right to the city. Lack of safety in both public spaces and public transport can restrict people's mobility. These impact vulnerable groups - the poor, elderly, women, children and disabled - deeply. Unsafe public spaces impede women's access to social and economic opportunities in life. Research has shown that both violence and fear of violence in public spaces including public transport instills a sense of fear in the mind of girls and women. A recent survey conducted by Observer Research Foundation in 2021, shows that 57 percent of women reported public transport in India to be unsafe, with a similar number - 56 percent, reporting that they have been sexually harassed while using public transport. Another conducted by Safetipin and FIA Foundation in 2020 says, sexual harassment in public places and public transport is quite common. 44% auditors (mostly girls and women in the neighbourhood) reported feeling unsafe at public places including streets, markets, bus stops.

The issue of women's safety in the city has come to the forefront primarily because of incidents of violence. Beyond the actual experience of violence though, the experience of the city for women and girls is often one of fear and exclusion. The issue needs to be seen as a form of discrimination faced by women and girls in cities which prevents their free movement and ability to access the city.

WOMEN IN PUBLIC SPACES OF CHENNAI

Sexual harassment against women and gender-based violence in public spaces and public transport affect gender equality in India. In Chennai similar issues of safety and accessibility adversely affects women's mobility, especially their ability to access employment, education, health care services, and leisure activities. According to 2019 National Crime Record Bureau (NCRB) data, Chennai has recorded fewer cases of crimes against women vis-a-vis other metro regions. However, there has been a slight increase in the number of cases between 2017 (642) and 2019 (729). In contrast to the low number of reported crimes, a study conducted by AWARE in Chennai found that more than 50 percent of women interviewed had faced some form of harassment while traveling on public transport.

According to the State Crime Records Bureau, under POSCO (The Protection of Children from Sexual Offence Act), 2,358 cases were registered in 2019, followed by molestation of women (803), cruelty by husbands and relatives (781), kidnapping and abduction (699) and rape (362). Cases booked under POSCO, molestation, cruelty by husband and his relatives, kidnapping and abduction together constitute 78.2% and rape accounted for 6.1% of the crime against women.

Nevertheless, we recognise that reported crime is probably only the tip of the iceberg and violence against women and girls is a much more pervasive phenomenon than demonstrated by crime statistics. It is only over the past decade that there has been an increased focus on public spaces and cities. Numerous research studies have been conducted to understand women's experience and participation in cities. In Delhi, Jagori has conducted several research studies and safety audits to better understand women's actual experience and response to urban situations. Research has also been carried out in Mumbai by Akshara and cities in Kerala by Sakhi which point to similar findings of fear and exclusion for women.

Lack of safety has adverse implications on the life of girls and women. These include lack of self-confidence, low productivity levels and most importantly it reduces their access to opportunities. To encourage girls and women to use public spaces freely, it must be made safe, inclusive and accessible particularly keeping the perspectives of women and girls in mind.

WOMEN SAFETY AUDIT BY SAFETIPIN

To address the safety concerns of women and girls in Chennai and to improve access, mobility and inclusion in the city, this project collected data on the quality, usage and perceptions of public spaces, particularly from a safety perspective of women and girls.

A Women's Safety Audit (WSA) was piloted in Tondiarpet, Chennai, using Safetipin digital tools to provide accurate data on aspects of the built environment which are linked to safety, accessibility and inclusion for women in public spaces. This information can be integrated into existing government GIS platforms and used to guide policies and interventions by the Government of Tamil Nadu (GoTN). WSAs have been used successfully globally, including in India, where data and evidence collected through such audits has led to a number of city-wide public infrastructure up-gradation projects which subsequently has improved the usage of spaces in the city.

Tondiarpet – George Town – Thiruvottiyur in Chennai was identified as the project area (18.75 Sq Km). It is located in the northern part of Chennai and is adjacent to other important regions of Chennai including George Town, which lies to the south. The region has its own railway station, Tondiarpet Railway Station, on the Chennai Central–Gummidipoondi Line. Trade and commerce along with manufacturing and distribution enterprises are predominant activities found at Tondiarpet.

Tondiarpet was identified for the following reasons as the project area.

1. It is one of the projects under the 'mega-street' program initiated by GCC aimed at redesigning and developing streets in Chennai to be future-ready and Non-Motorised Transport friendly incorporating various aspects of mobility, utility and liveability over 2021 to 2023.
2. Various areas in this location have been selected under the Nirbhaya Fund for installing safety poles, aimed at making areas safer for women.

3. North Chennai is considered relatively unsafe compared to the rest of the city.
4. The area has connectivity with public buses, sub-urban rail and a new metro line.

The focused study of Tondiarpet will help guide the existing projects and propose recommendations for implementing additional projects.

UNESCO's definition of public space refers to an area or place that is open and accessible to all people, regardless of gender, race, ethnicity, age or socio-economic level. These are public gathering spaces such as plazas, squares and parks. Spaces connecting these, such as footpaths and streets, are also public spaces. Within such reference, the study collected data mainly on the streets and footpaths to evaluate safety, accessibility and inclusion; conducted analysis of the collected data and gave recommendations to improve streets and footpaths. In addition, this project intended to provide analysed data to city stakeholders for decision making in order to make city spaces accessible, safe and inclusive by responding to the priority issues identified through the use of the Safetipin application.

The 'women safety audit of public spaces' was conducted using two application - Safetipin Nite and My safetipin. The study mapped through images of all the main roads of Tondiarpet. The app provided large scale data at the street level to investigate on the challenges on safety and accessibility encountered by women while using public spaces in the area. Further it conducted analysis of the data and highlighted the issues around safety, accessibility and inclusion and provided recommendations to improve women's access to and participation in public spaces, and to meet the current and future demands of the area exclusively for public space usability. This data could be valuable source of information for both citizens and governments. The details of how the applications were used and data was produced to give recommendations for improvement are given in the following chapters.

SAFETY AUDIT METHODOLOGY

Safety mapping using Safetipin applications

IMAGE MAPPING WITH SAFETIPIN NITE

Two Safetipin applications are used for this project - Safetipin Nite and My Safetipin. Safetipin Nite application generates data by clicking pictures across the city via a phone camera mounted on the windshield of a moving vehicle. Photos are taken to capture pedestrian's safety and accessibility conditions at regular intervals of 30-50 metres. Every picture is geo-tagged and uploaded on to a server and further analysed by a team of expert coders using Safetipin parameters and sub-parameters. These eventually are translated into data sets that appear as audit points on Safetipin maps. This image bank of GIS data is then shared with local stakeholders, government authorities and concerned organisations in order to address and improve safety and accessibility conditions in the city.

PUBLIC SPACE AUDITS WITH MY SAFETIPIN

My Safetipin application has the Women's Safety Audit at its core. Based on the global experience of 20 years on safety audits, 8 important parameters namely, lighting, walk path, public transport, visibility, openness, crowd, gender diversity and security that define safety perceptions are fixed. Feeling being the ninth parameter is exclusively on perception. These together help to understand the perception of safety of public spaces. A location is audited based on these nine parameters. Each safety audit results in a pin on the specific geo-tagged location where the audit was performed and also records the time and date. Results of the safety audit are aggregated to a 'Safety Score' for areas and neighbourhoods. The information on the application specifically focuses on the experience of women and girls, providing them with an interactive and empowering tool around feelings of safety.

SAFETIPIN AUDIT PARAMETERS TO EVALUATE SAFETY AND ACCESSIBILITY IN PUBLIC SPACES:

Lighting – Availability of enough light to see all around you

Walkpath – Either a pavement or road with space to walk

Public Transport – Availability of public transport like metro, buses, autos, rickshaws

Visibility – Vendors, shops, building entrances, windows and balconies from where you can be seen

Security – Presence of formal police or private guards

People – Number of people around you

Gender Usage – Presence of women and children near you

Openness – Ability to see clearly and move in all directions

Feeling – How safe do you feel

Parameter	Score 0	Score 1	Score 2	Score 3
Lighting (Night)	None	Little	Enough	Bright
	No street or other lights	Can see lights, but there is low visibility in the area	Lighting is enough for clear visibility	Whole area brightly lit
Openness	Not Open	Partly Open	Mostly Open	Completely Open
	Many blind corners and no clear sightlines	Able to see a little ahead and around	Able to see in most directions	Can see clearly in all directions
Visibility	No Eyes	Few Eyes	More Eyes	Highly Visible
	No windows or entrances of shops or residences overlook the point	Less than 5 windows or entrances overlook the point	Less than 10 windows or entrances and vendors overlook the point	More than 10 windows or entrances and vendors overlook this point
People	Deserted	Few People	Some Crowd	Crowded
	No one in sight	Less than 10 people in sight	More than 10 people visible	Many people within touching distance
Security	None	Minimal	Moderate	High
	No guards or police visible in surrounding area	Some private security visible in surrounding area but not nearby	Private security within hailing distance	Police / reliable security within hailing distance
Walk Path	None	Poor	Fair	Good
	No walking path available	Path exists but in very bad condition	Can walk but not run	Easy to walk fast or run
Public Transport	Unavailable	Distant	Nearby	Very Close
	No metro, bus, auto/rickshaw stop within 10 mins walk	Metro or bus auto/rickshaw stop between 5-10 mins walk	Metro or bus, auto/rickshaw stop between 2-5 mins walk	Metro or bus, auto/rickshaw available within 2 mins walk
Gender Usage	Not Diverse	Somewhat Diverse	Fairly Diverse	Diverse
	No one in sight, or only men	Mostly men, very few women or children	Some women and children	Balance of all genders or more women and children
Feeling	Frightening	Uncomfortable	Acceptable	Comfortable
	Not venture here without sufficient escort	Will avoid this place whenever possible	Feel safe enough, but will be careful	Feel safe here even after dark

THE RATING RUBRIC

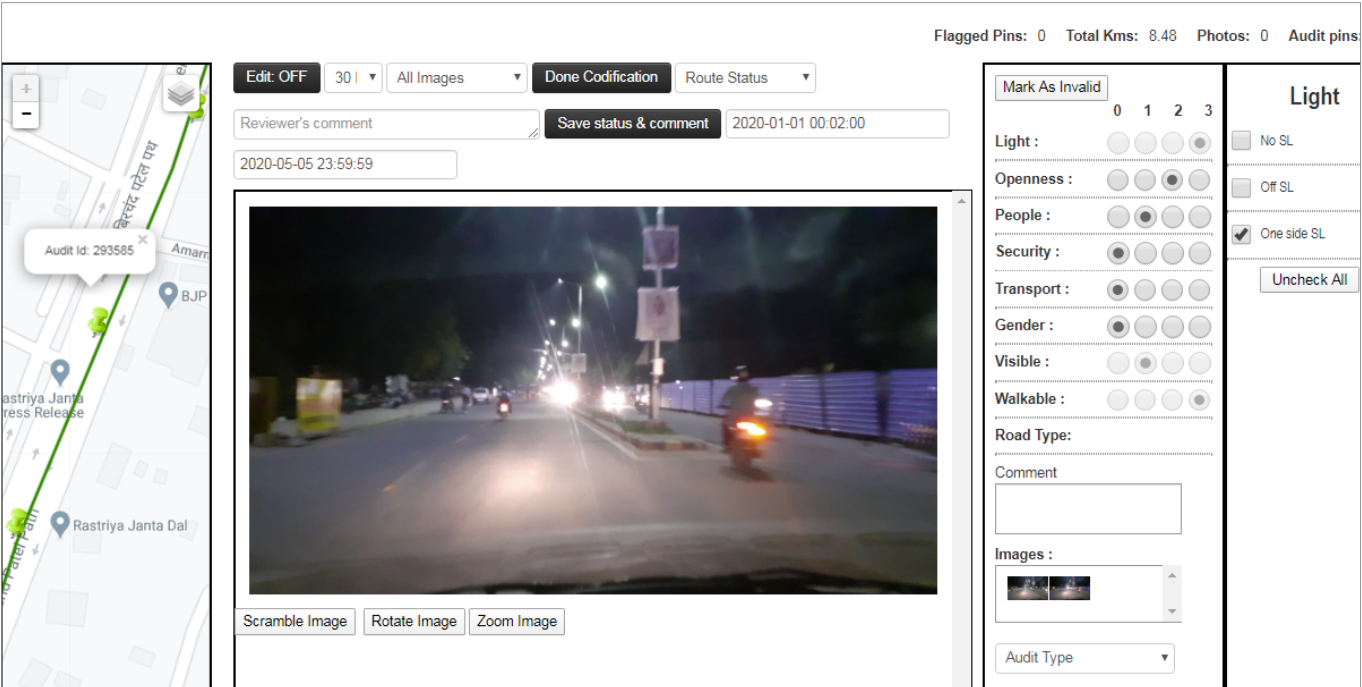
Image on left
The ratings for each of the parameters are defined on a scale of 0-3

ACCESSIBILITY MAPPING WITH SAFETIPIN NITE

To gain a deeper understanding of accessibility and safety in public spaces of Tondiarpet, a thorough investigation on accessibility was done. Safetipin Nite was used to map the area twice - during the day and after dark. An additional set of accessibility parameters apart from the standard Safetipin parameters explained before were applied to assess the status of physical accessibility in the area especially from a user's perspective. Since walking is a basic and common means of access in the city, walkpaths which facilitate walking, become the crucial form of urban infrastructure which provides physical access to services and resources. Day and night image mapping of walkpaths were put together to analyse the ease of accessibility in the area. The additional accessibility parameters applied are as follows:

- Footpath Conditions:
- Non-existent footpath
- Broken, unpaved, dis-continuous footpath
- Availability of curb ramps, median breaks and zebra crossings
- Footpath Barriers:
- Garbage, construction debris/material dumped on footpaths
- Vehicles parked on footpaths
- House and shop extensions onto footpaths
- Temporary stalls and vendors obstructing footpaths
- Light, electric, signage and advert poles obstructing footpath

Image below
Snapshot of the coding screen showing the parameters and sub-parameters



DATA CODING ON SAFETIPIN PORTAL

The photographs collected through Safetipin Nite are uploaded onto the Safetipin server. These are then accessed on the portal and analysed on a wide range of parameters linked to accessibility, safety and inclusivity of public spaces. For deeper analysis, a set of sub-parametres for the parameters of Lighting, Walkpath, Public Transport, Visibility and Security are used for recording additional information. The complete list of sub-parameters is given here. Google Maps are also used to identify the sub-parameters. Every 100 metres, an audit is generated at a point by using all the photographs collected around that point.

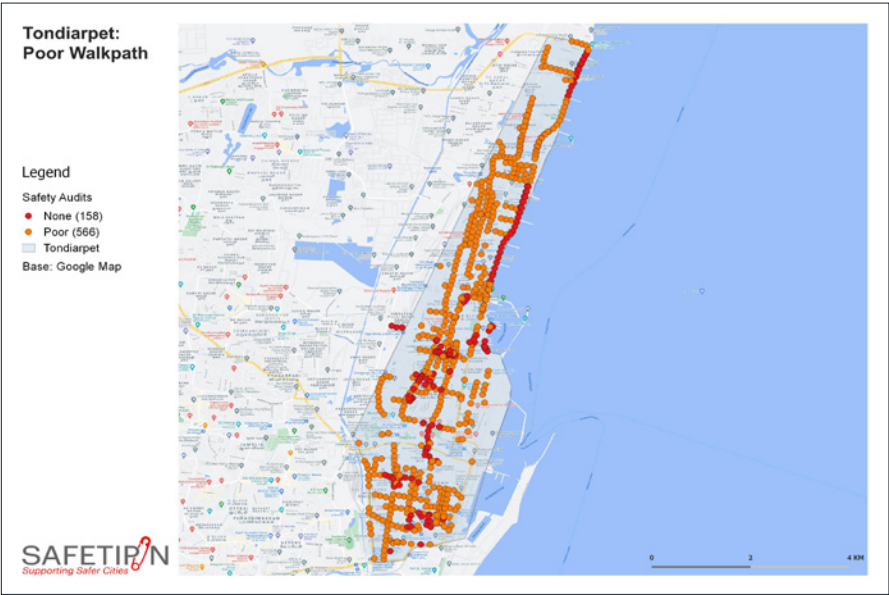
GENERATING MAPS, ANALYSES AND REPORTS

Based on the collected and coded data, the Safetipin team analyses and produces maps and reports that can be used for actual on ground responses and actions to work toward improving safety and accessibility on the streets and public spaces in cities. Maps are produced on each parameter with geo-tagged location of audit points, supported with images. These maps could be seen on Google Maps too for easier assess to information. Data is also analysed in terms of correlations and linkages with other possible data sets which further supports or reinforce the findings. Based on these analyses, a concrete set of recommendations are drawn out for key stakeholders for city improvement programs.

Lighting	Visibility	Walkpath	Security	Public Transport
No SL	50% Boundary Wall	No Pavement	Private Guards	Metro/Rail
Off SL	100% Boundary Wall	Broken Pavement	Police Van/Bike	Bus/Mini bus
Dim SL	Unused Land	Unpaved Pavement	Police Check	Auto/Shared Auto
High SL	Road side Vendors	Car Blocking	Police Booth	Cycle Rickshaw
Too Far SL	Temporary Stalls	Vendor Blocking	Police Station	Taxi
Leaves cover SL	Shops	Houses Extending	Other Govt.	Rental Bicycle
Other cover SL	Houses upto 4 Floors	Trees Blocking		
One side SL	Houses >4 Floors	Other Blocking		
		Walkable Road		
		On-street Parking		

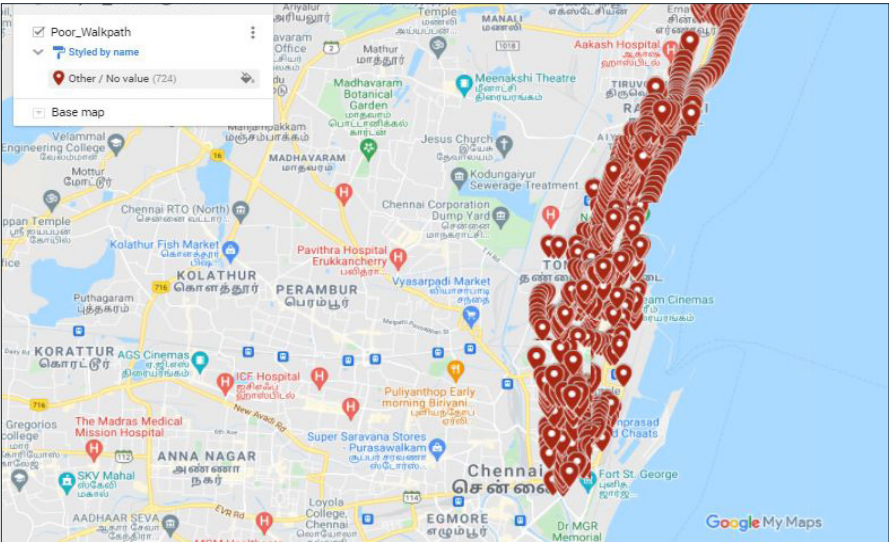
SUB-PARAMETER RUBRIC

Image on left
Rubric showing all the sub-parameters for the listed five parameters



TONDIARPET'S POOR WALKPATH AS AUDIT MAP

Image on left
The map shows locations in the city where there is no or poor street lighting



TONDIARPET'S POOR WALKPATH AS GIS LAYER

Image on left
GIS layer of street lighting on Google Maps with image links

DATA COLLECTION

Data collection using Safetipin Applications

DATA COLLECTORS ONLINE APP TRAINING

As soon as the lockdown and night curfew due to the second wave Covid crisis in Chennai was lifted and vehicular movement was allowed in the city after dark, Safetipin conducted ‘Online Data Collectors’ Training Program’ with the verified drivers and field investigators assigned by a resource agency in Chennai. The agency - Market Xcel hired by Safetipin supported the Safetipin team to individually train every driver and field investigators through online sessions given by Safetipin team to use ‘Safetipin Nite’ application on the smart phones provided by Safetipin.

Over the app training sessions, Safetipin Nite App was introduced to two field investigators and two drivers who were chosen for the work. They were even trained to use the app in their personal phones as backups. Alongside, they were also trained on how to mount the phones on the windshields of the cars to do the image collection. Pre-assigned data collection routes were run on their phones and few rounds of test runs were done with the field investigators and drivers to check the accuracy of the image collection process.

Data collection details:

- Total area covered: 19.3 Sq Km
- Total road length covered: 321 Km
- Image collection time:
- Day - 3:00 pm to 5:30 pm, Night - 6:30 pm to 9:00 pm
- Total images collected: 20070 nos
- Total audits points - 1098

DATA COLLECTION WITH SAFETIPIN NITE

A distance of 321 km were covered by the drivers and their respective field investigators in Tondiarpet, collecting 20070 images of the area between 6:30pm to 9:00pm. Similarly, the area was also mapped during the day between 3:00pm to 5:30pm for mapping accessibility conditions. The quality of most of the images collected were certified good after an initial round of review by the Safetipin team and were processed for the main visual analysis. The field investigators and drivers worked efficiently in collecting data with the help of Safetipin Nite app.

Nevertheless, we met with a couple of challenges. These are as follows:

On certain routes because of the road width being insufficient for vehicles it was difficult accessing them.

We received a small percentage of shaky images due to bad road condition, water logging in monsoons and rain water blurring the camera view.



DATA COLLECTION CAR

Image on left
Safetipin Nite app running on volunteer's and driver's smart phones



GOOD IMAGE

Image on left
Photo collected through Safetipin Nite app - an example of a good image



BAD IMAGE

Image on left
Photo collected through Safetipin Nite app - an example of a bad image

DATA COLLECTION

Data collection using Safetipin Applications

PUBLIC SPACE AUDITS WITH MY SAFETIPIN

Two women data collectors were selected by the resource agency for conducting public space safety audits. Safetipin team trained both the women data collectors through two online 'App Training' sessions. The first session of the training initiated informal discussions with the data collectors on women's safety in public spaces and the significance of safety audits. The second session was conducted on the user generated audits using My Safetipin application.

During the second session, the women data collectors were introduced with the functioning of the My Safetipin app and were helped to install the app in their smart phones. After installing the app each data collector did some trial audits to make sure that they have understood the process correctly. After trial audits were done, each volunteer was assigned with areas which were convenient for them to access and do the safety audits. These two sessions of the training complemented each other in generating quality data at the end of the audits.

13 public places comprising of transport stations, markets, schools, colleges and parks were audited. Each audit place generated 10-15 audit points in and around the identified locations.

While we were able to collect good quality data but due to the ongoing pandemic, Safetipin team had to re-train a second batch of data collectors as the first one could not perform the expected work. This duplication of work extended the data collection process. Women auditors at few instances were not able to audit public spaces at night.

Public spaces audited with My Safetipin app:

Railway Station - Chennai Fort

Metro Station - New Washermenpet Station

Bus Stop - Broadway Bus Stop

School - St Annes Girls' Higher Secondary School

College
Ibrahim Salai Bharathi Arts College
Kanniyaga Parameswari Women's College

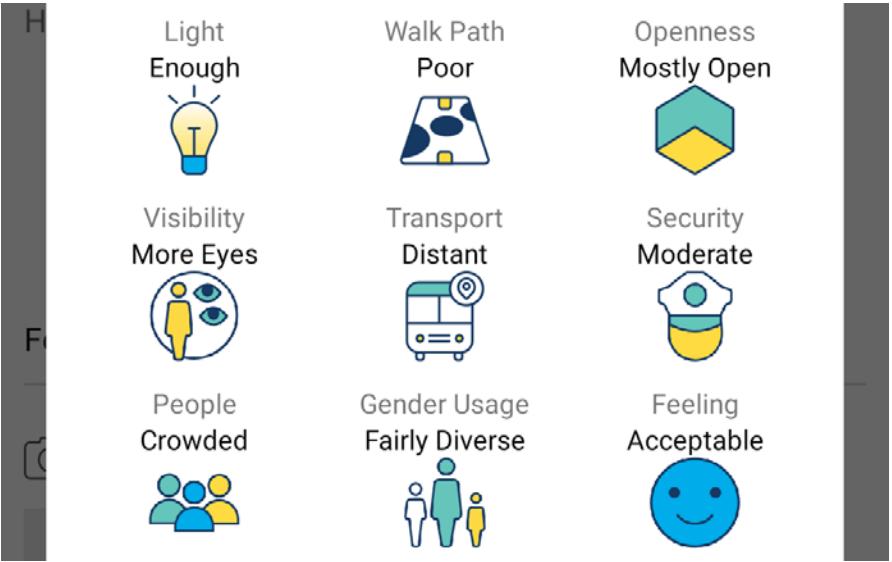
Hostel - Arts College Hostel

Park - Thiyagaraya Park

Market
Tondiarpet Market
Periya/Washermanpet Market
Kasimedu Fish Market

Hospital - Mint Street Apollo Hospital

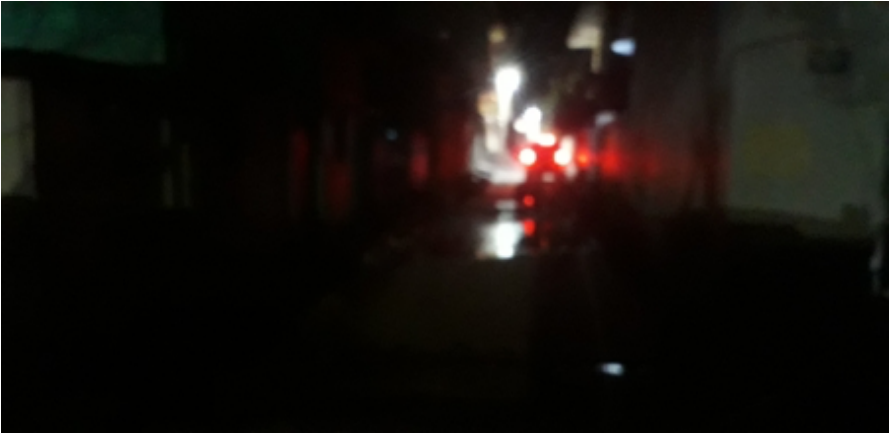
Public Building - High Court



MY SAFETIPIN SCREEN

Image on left

My Safetipin app screen showing all the nine audit parameters



BAD IMAGE

Image on left

Photo collected through My Safetipin app - an example of a bad image



GOOD IMAGE

Image on left

Photo collected through My Safetipin app - an example of a good image

WOMEN'S SAFETY AUDIT FINDINGS

Parameter maps and safety ratings

After analysis of the collected data by Safetipin Nite in Tondiarpet, representational maps for six key parameters were prepared for data visualisation. The following parameter maps represent the data as audit points generated through image analysis technique. The parameter maps also include user generated audits points by My Safetipin. An overall map showing the aggregated Safety Score of all the parameters combined is generated for safety rating for the audited locations in Tondiarpet. These maps help to read and compare the data geographically and understand where the gap lies in making public spaces safe and accessible, especially for women and others.

Overall, 1098 audit points were generated for Tondiarpet. Geo-located Safety Audit points by Safetipin Nite - 919 and by My Safetipin - 179.

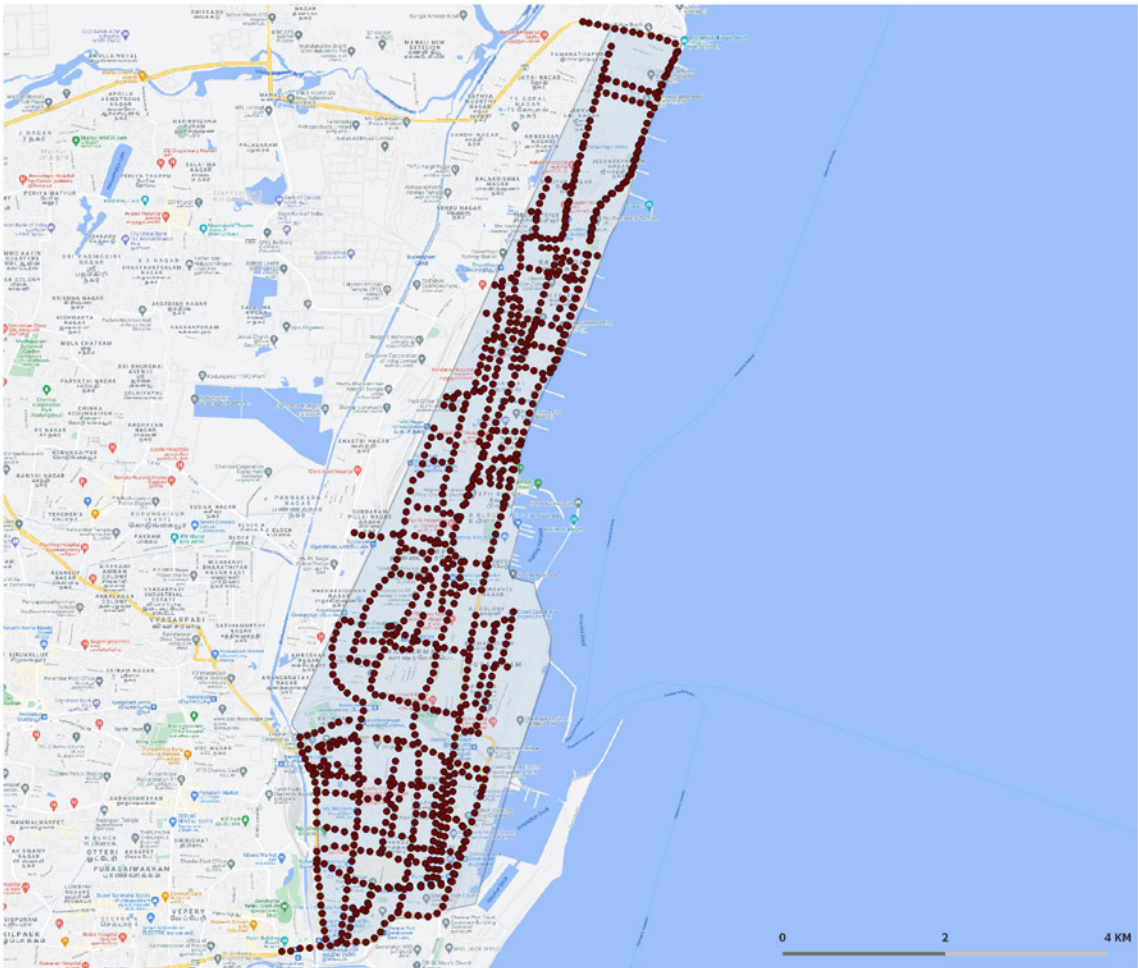
The map below shows the safety audit points mapped by Safetipin Nite application.

Chennai: Safetipin Nite Audits

Legend

- Safety Audits (919)
- Chennai

Base: Google Map

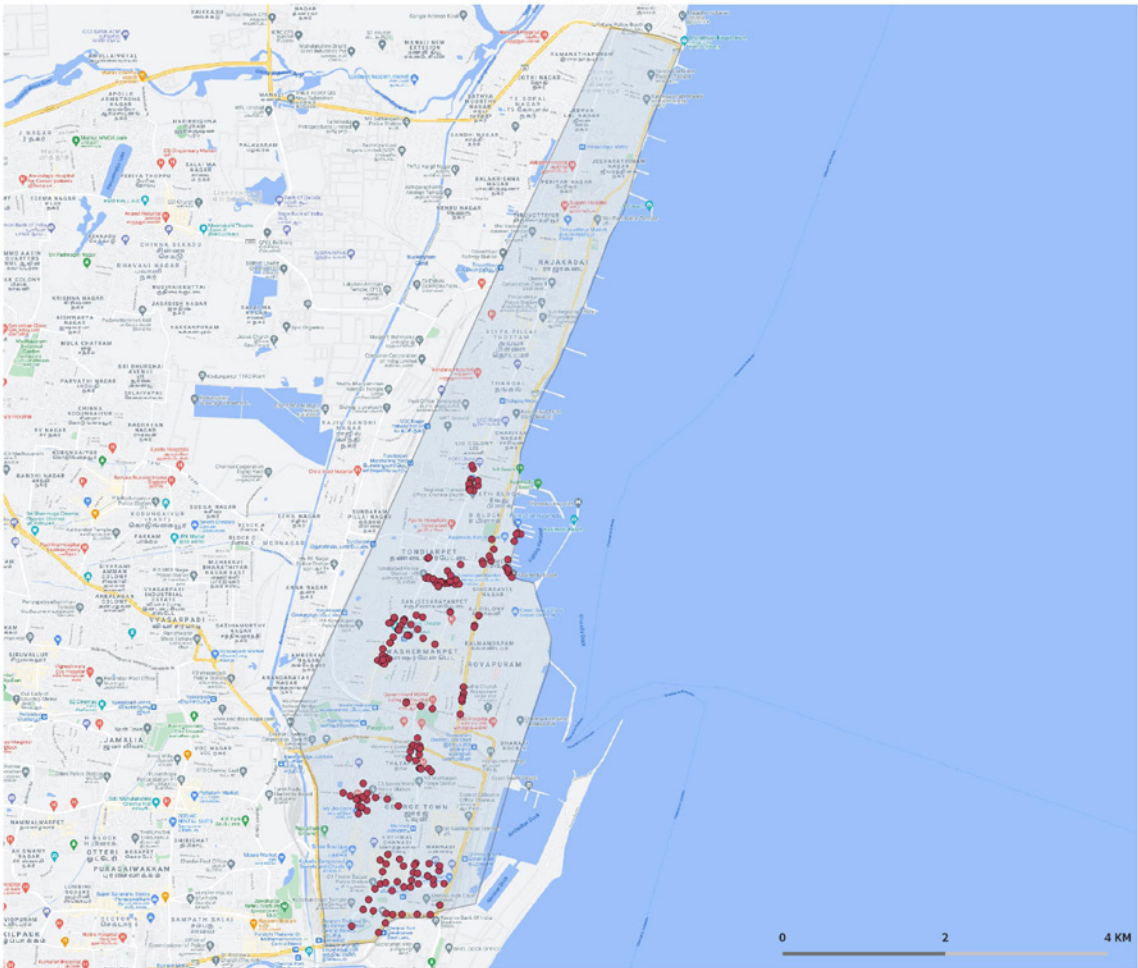


Chennai: My Safetipin Audits

Legend

- Safety Audits (179)
- Chennai

Base: Google Map



User generated public space audits with My Safetipin app:

Two women data collectors audited the selected public spaces in Tondiarpet. As an additional layer of information they included comments for some of their audits. An assortment of comments with user insights are given below for analysis

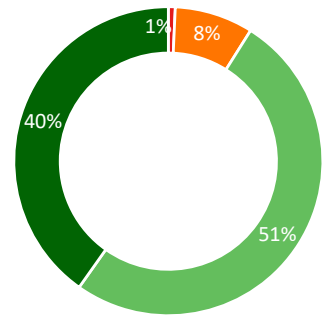
- "No lights nothing, very afraid to walk in the streets."
- "Women feel uncomfortable because no lighting facility there."
- "Some space is there in the road side but no proper walk path."
- "Crowded area always full of vehicles, no space to walk, may meet with an accident because of no space."
- "No vehicles, autos nearby, need walk for 10 mins."
- "Shops closed no female at all."
- "Police booth were there but no police presence."

Of the nine parameters used for mapping and evaluation of public space, Lighting, Walkpath, Public Transport and Visibility are actionable parameters on which physical interventions are possible. People (presence of people on the streets), Gender Diversity (presence of women, children and others in the public domain) and Feeling (perception of safety) are used to measure the impact of the actionable parameters on safety, accessibility and inclusion in public spaces. Security and Openness are parameters which provide supplementary information for overall analysis. Findings on the actionable parameters Lighting, Walkpath, Public Transport, Visibility are given below along with People and Gender Diversity.

The map below shows the safety audit points mapped by My Safetipin application.

91%

OF THE AREA MAPPED HAS
ADEQUATE STREETLIGHTS



91% of the area has adequate street lighting and only 9% has not any or poor street lighting. When this data is represented geographically (map below) it is seen that areas with lower ratings on street lighting are sporadically spread across at few locations. This means, either there are not any or non-functional street lighting in these parts of the mapped area. This geo-located data, provided as a GIS layer, could be used for improving street lighting in the area and making it fully functional.

On the left, percentage distribution pie of the parameter map below. Tally with the legend below.

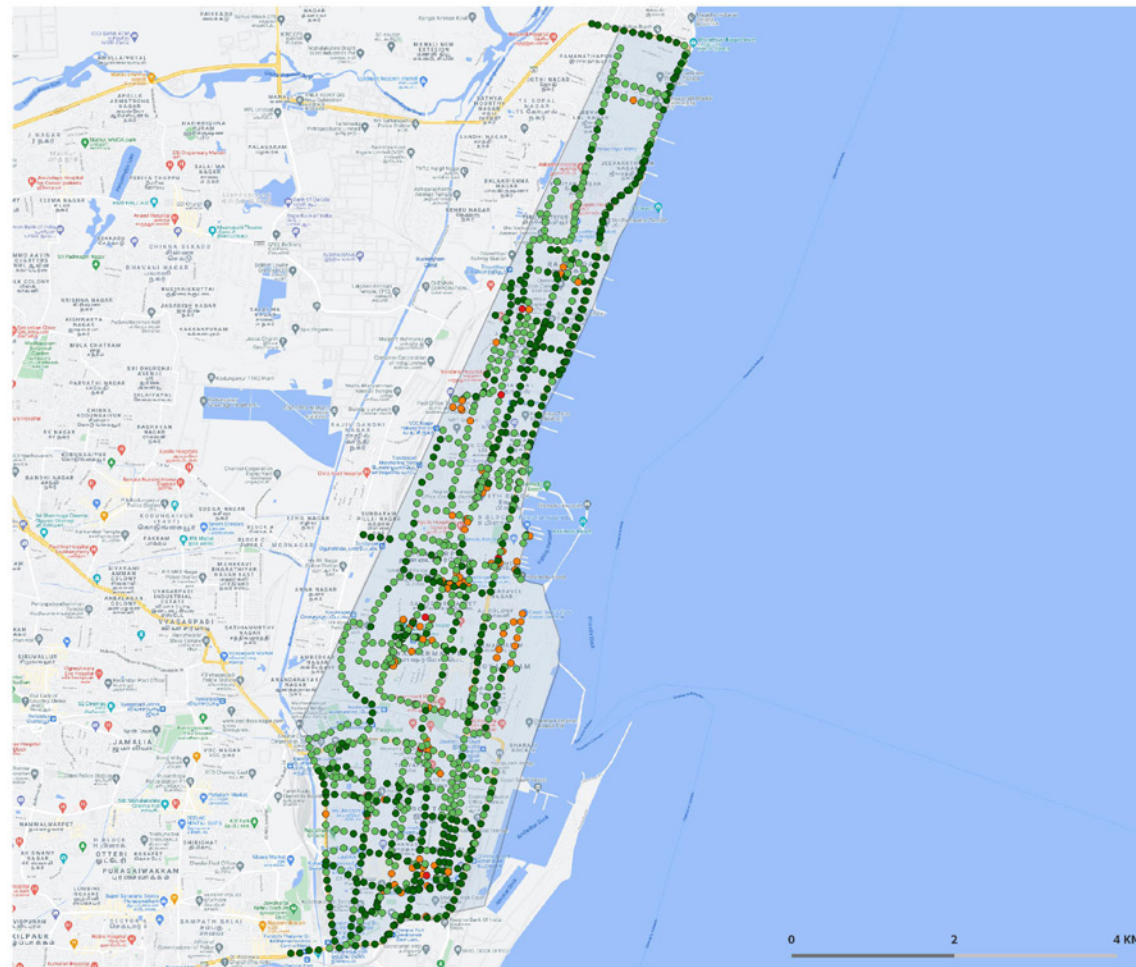
Chennai: Parameter Lighting

Legend

Safety Audits

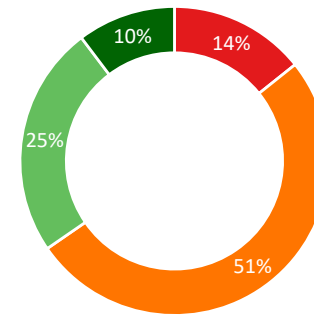
- Poor Light
- Some Light
- Enough Light
- Bright Light
- Chennai

Base: Google Map



35%

OF THE AREA MAPPED HAS
WALKABLE FOOTPATHS



35% of the area has walkable footpaths, rest 65% scores poorly on the same. When this data is represented geographically (map below) it is seen that areas with lower ratings on footpaths are spread across the mapped area with an exception of a small part at the southern end and along few corridors leading north. This means, either there are not any or very poor quality footpaths in these parts of the city. A detailed footpath analysis follows to understand accessibility better.

On the left, percentage distribution pie of the parameter map below. Tally with the legend below.

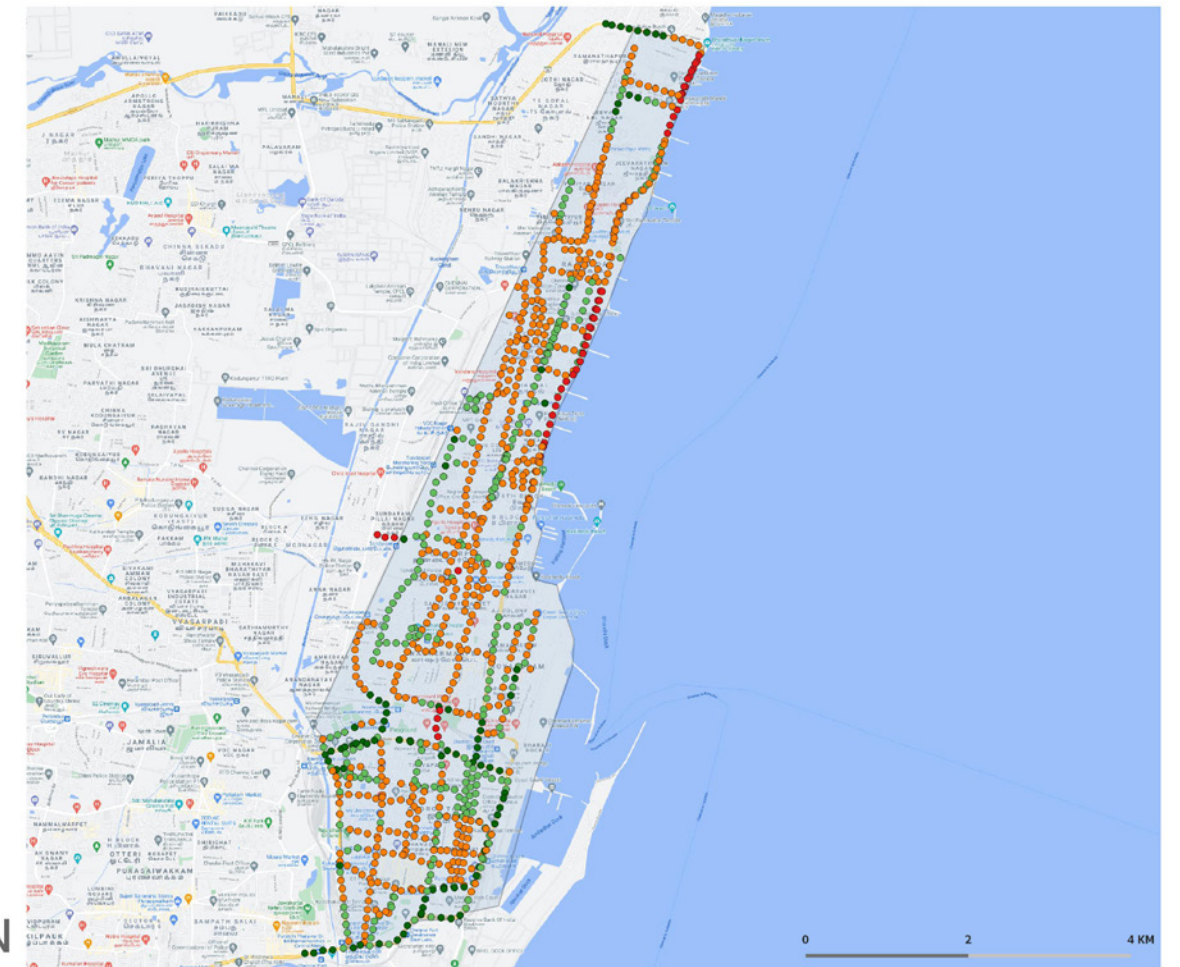
Chennai: Parameter Walkpath

Legend

Safety Audits

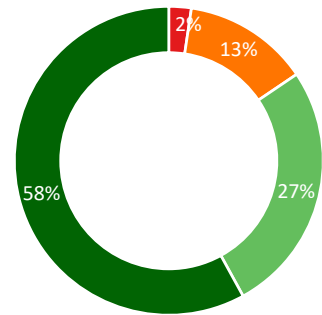
- None
- Poor
- Fair
- Good
- Chennai

Base: Google Map



85%

OF THE AREA MAPPED HAS PUBLIC TRANSPORT STOPS WITHIN REACH



58% of the area has public transport stands/stops within 2-5 minutes walking distance, 27% can reach them in 5-10 minutes and the rest 15% does not have any formal or informal transport stands/stops reachable within 10 minutes. When this data is represented geographically (map below) it is seen that areas with lower ratings on public transport are in a clusters in the south and in small clusters at few more locations moving north. This means, formal public transport stands/stops are either unavailable or not within reach in these parts.

On the left, percentage distribution pie of the parameter map below. Tally with the legend below.

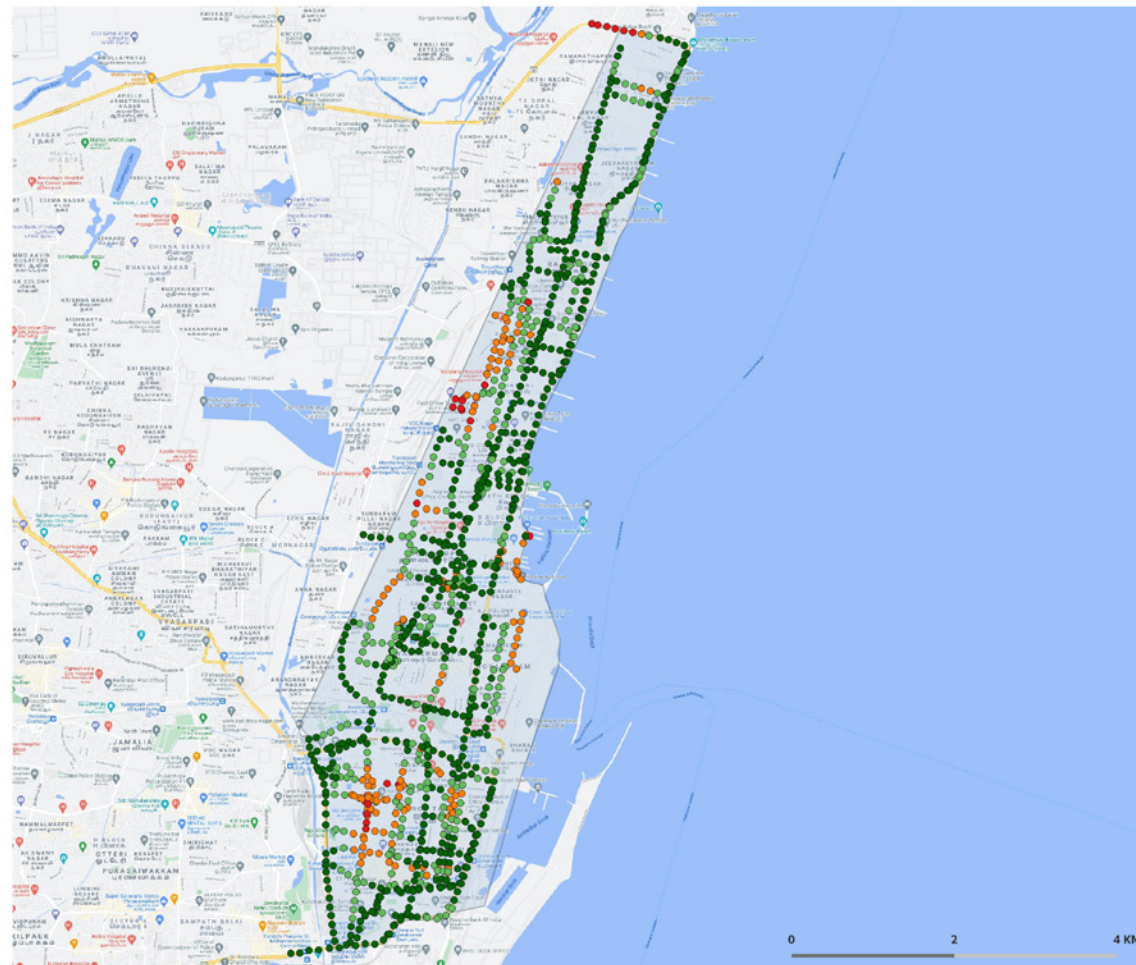
Chennai: Parameter Public Transport

Legend

Safety Audits

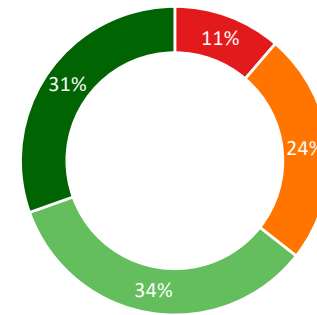
- Unavailable
- Distant
- Nearby
- Very Close
- Chennai

Base: Google Map



65%

OF THE AREA MAPPED HAS GOOD VISIBILITY



65% of the area has good visibility or has active streets, i.e., these parts have less number of high boundary walls or blank facades facing the streets and have more people, vendors, shops on the streets. But 35% of the area still scores poorly on the same. When this data is represented geographically (map below) it is seen that areas with lower ratings on visibility are spread across the mapped area with some long stretches along a western corridor by the sea in the north. This means, either there are not any or very low visibility.

On the left, percentage distribution pie of the parameter map below. Tally with the legend below.

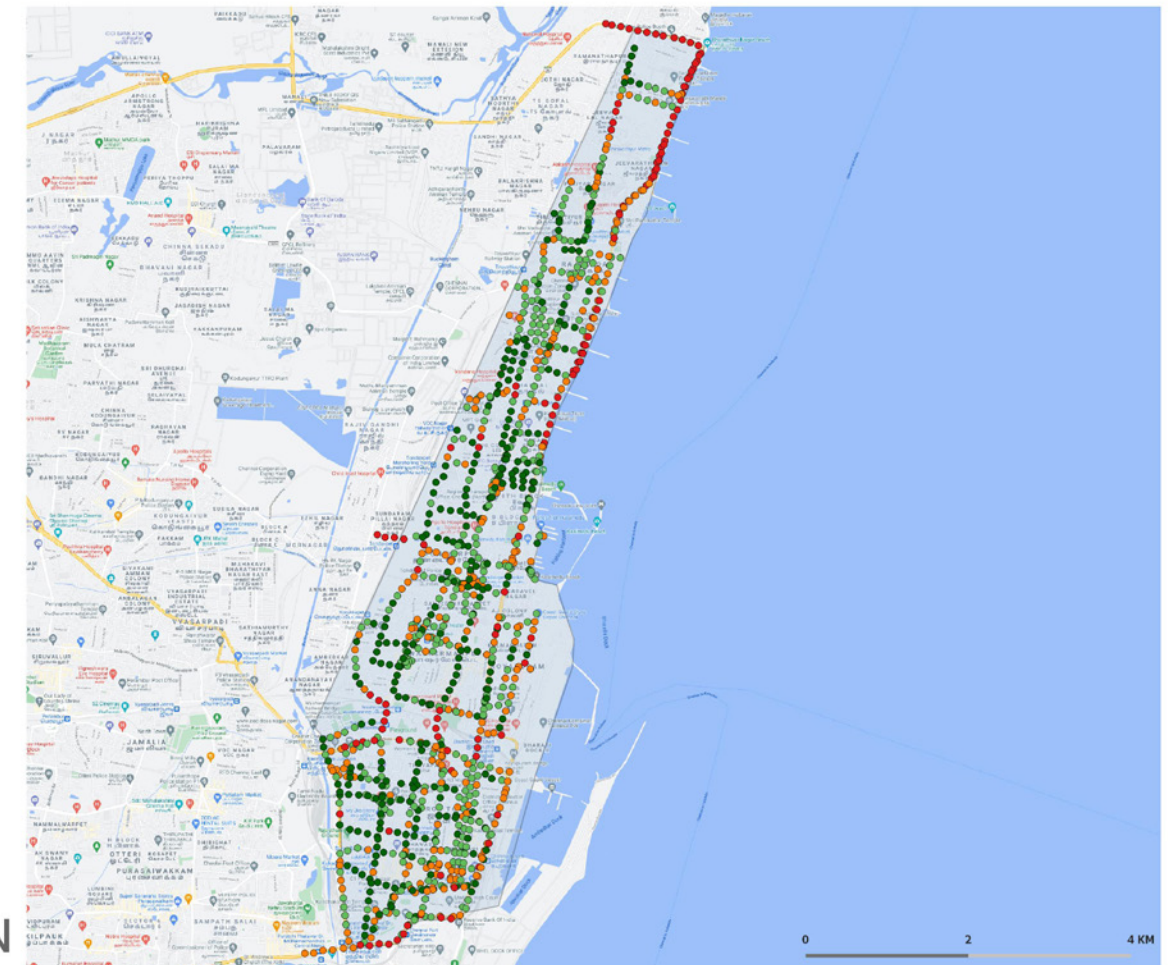
Chennai: Parameter Visibility

Legend

Safety Audits

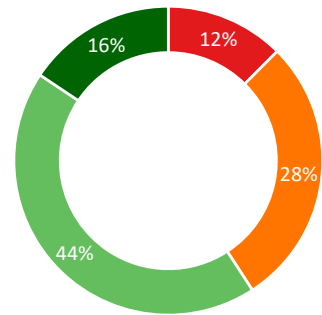
- No Eyes
- Few Eyes
- More Eyes
- Highly Visible
- Chennai

Base: Google Map



60%

OF THE AREA MAPPED HAS GOOD PRESENCE OF PEOPLE



60% of the area had good number of people present on the streets during the time of the audit, while 28% had less than 10 people after dark and 12% were quite empty. When this data is represented geographically (map below) it is seen that areas with lower ratings on people are spread across the mapped area at various locations. The western most corridor by the sea is particularly deserted in the north. This means, either there were no or few people in these parts during the mapping. This data could be supplemented with other data sets to understand why these parts are underused.

On the left, percentage distribution pie of the parameter map below. Tally with the legend below.

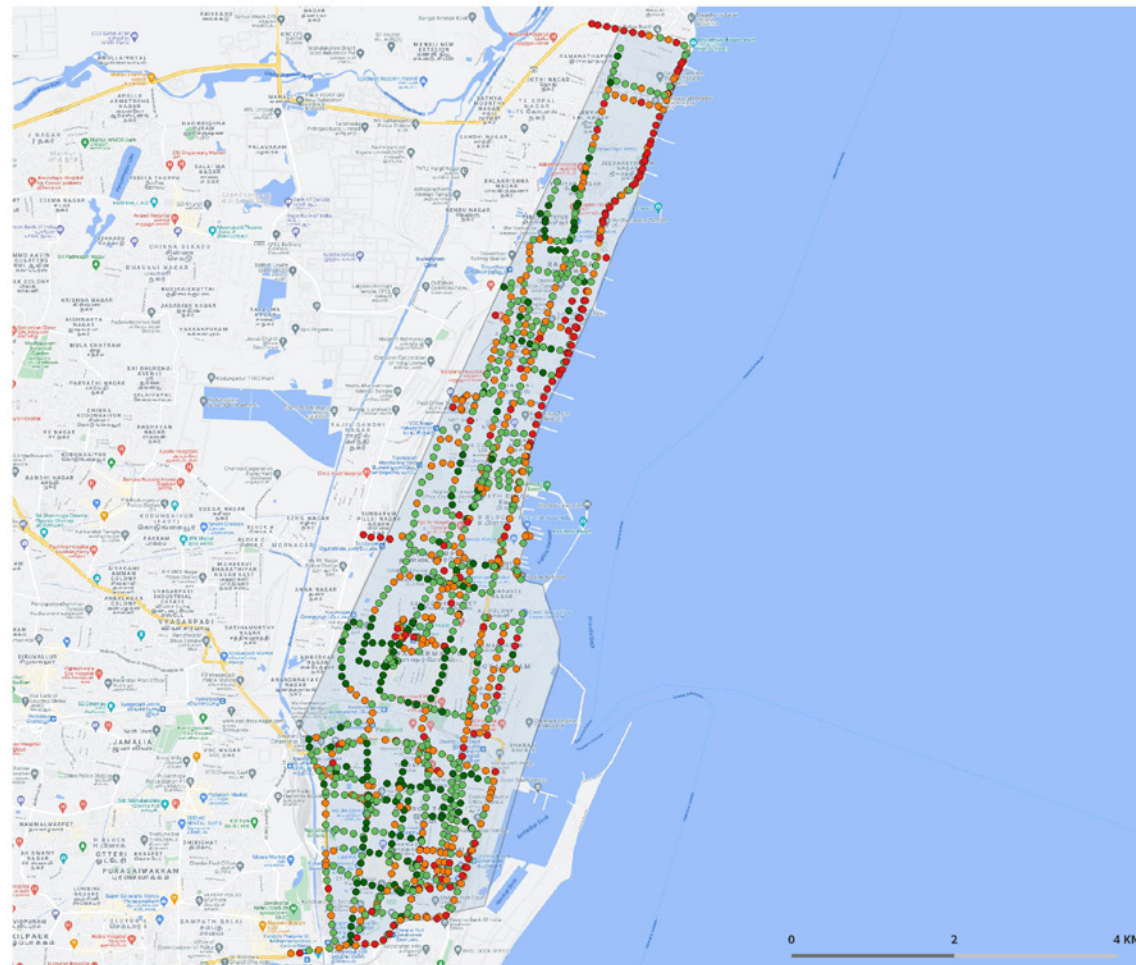
Chennai: Parameter People

Legend

Safety Audits

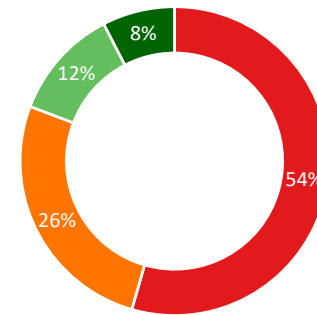
- Deserted
- Few People
- Some Crowd
- Crowded
- Chennai

Base: Google Map



20%

OF THE AREA MAPPED HAS PRESENCE OF WOMEN



Only 20% of the area had women and children present in public spaces during the time of the audit, rest 80% had either no or very few women and children present on the streets. When this data is represented geographically (map below) it is seen that areas with lower ratings on gender usage are evenly present across the entire area barring few locations. This means, that there were no women and children present at night on most of the streets in the mapped area. This data too could be supplemented with other data sets to understand why streets are underused by women and children.

On the left, percentage distribution pie of the parameter map below. Tally with the legend below.

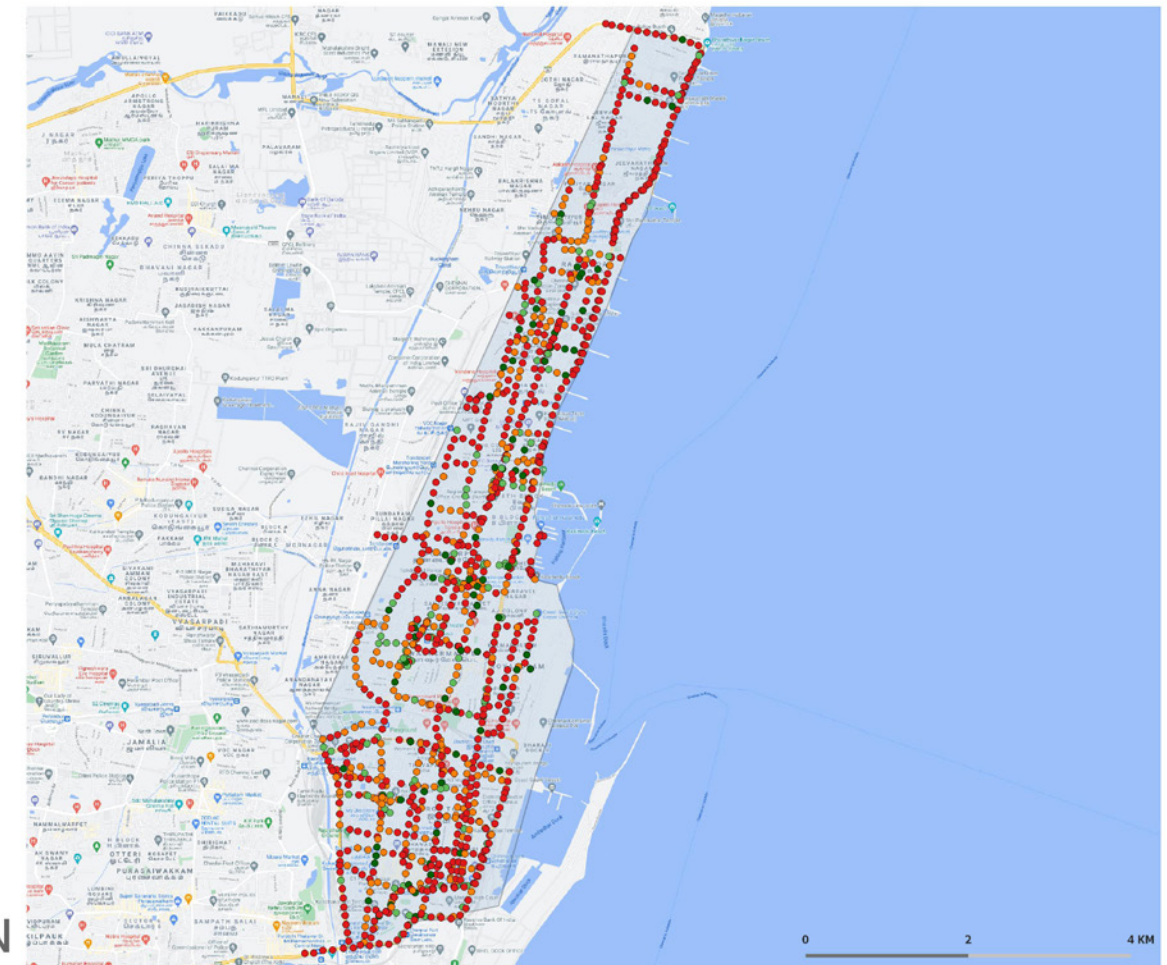
Chennai: Parameter Gender Usage

Legend

Safety Audits

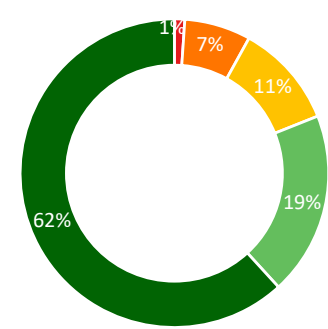
- Not Diverse
- Somewhat Diverse
- Fairly Diverse
- Diverse
- Chennai

Base: Google Map



3.8/5

SAFETY SCORE OF TONDIARPET



Based on the collected data, a safety score of 3.8 on a scale of 5 is given to Tondiarpet which evaluates its status as 'above average' on safety and accessibility. The safety score map below shows some areas across Tondiarpet with poor physical infrastructure and poor usage of public space. This means that the condition of street-lighting, footpath, visibility and public transport needs improvement at these locations and would together contribute to make Tondiarpet safe and accessible for women and others. A thorough analysis follows to understand which of the parameters must be prioritised to improve the overall safety status of the mapped area.

On the left, percentage distribution pie of the safety score map below. Tally with the legend below.

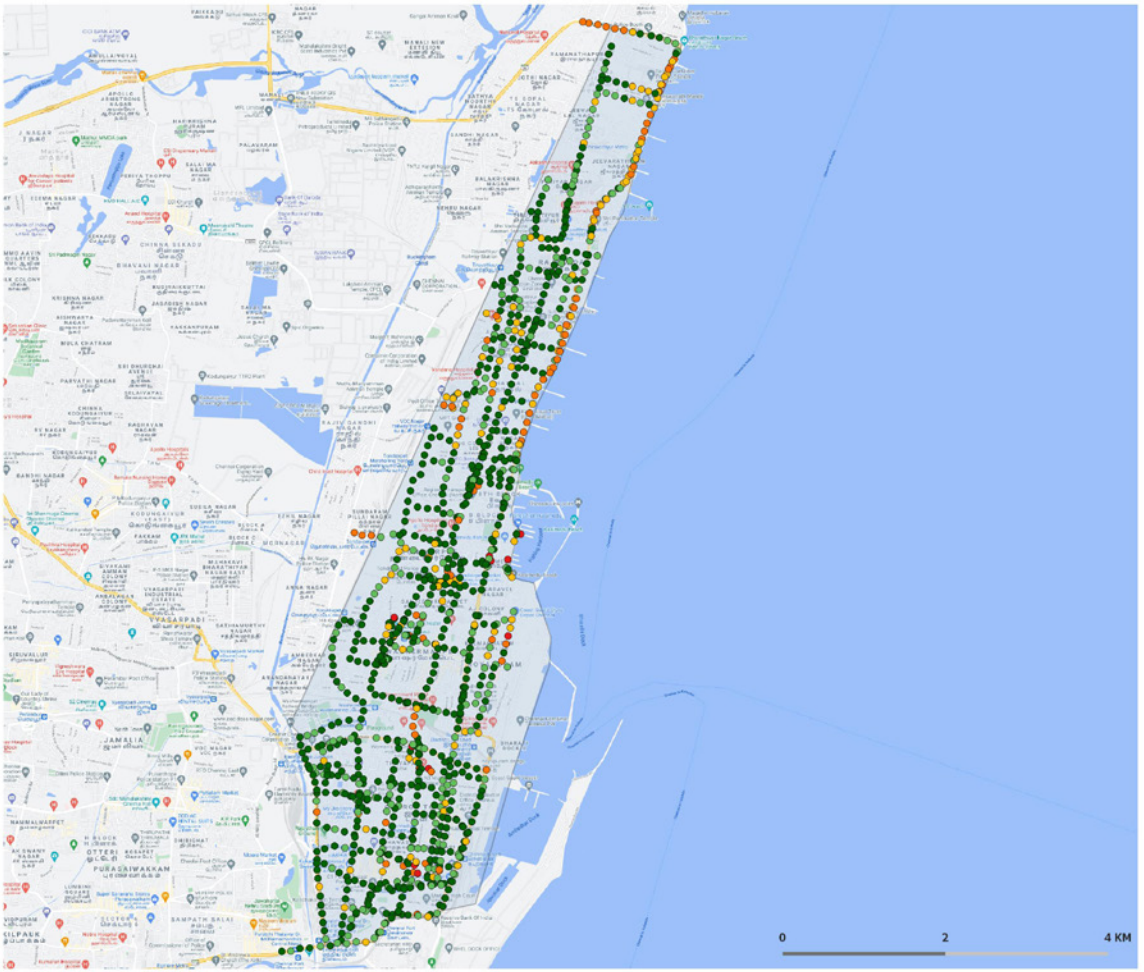
Chennai: Safety Score

Legend

Safety Audits

- Poor
- Below Average
- Average
- Above Average
- Good
- Chennai

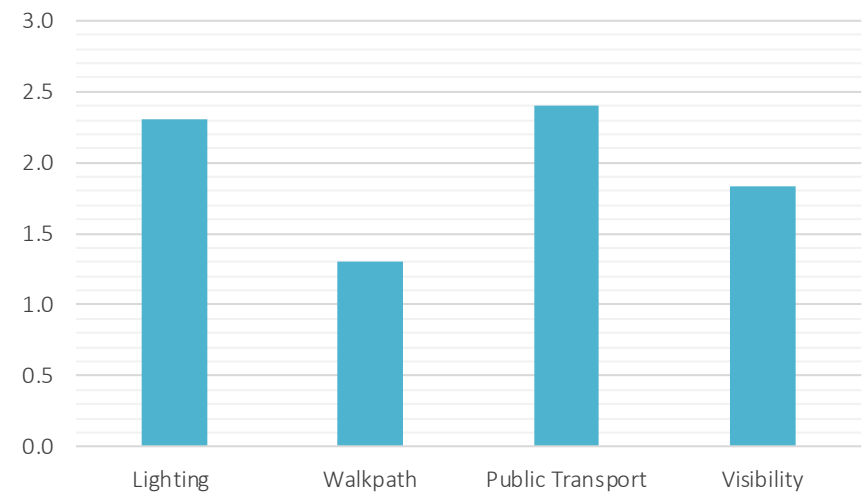
Base: Google Map



GRAPHIC INTERPRETATION OF KEY PARAMETERS

To further analyse, the parameters on which physical interventions are possible are put together and the data is compared. The average parameter rating graph indicates the average rating for each parameter on a scale of three. Each of the four parameters are rated either 0,1, 2 or 3, where 0 is poor and 3 is good. As seen on the graph, the Public Transport parameter has been rated the highest, followed by other parameters such as Lighting and Visibility which are rated little lower. Walkpath has been rated the lowest. The parameter wise pin distribution graph indicates the number of points rated as 0, 1, 2 and 3. The good ratings are taken as positive and poor ratings as negative. As shown on the graph, for Walkpath more than half audit points are rated below average or poor. Whereas, for Public Transport and Visibility most audit points are rated good or average.

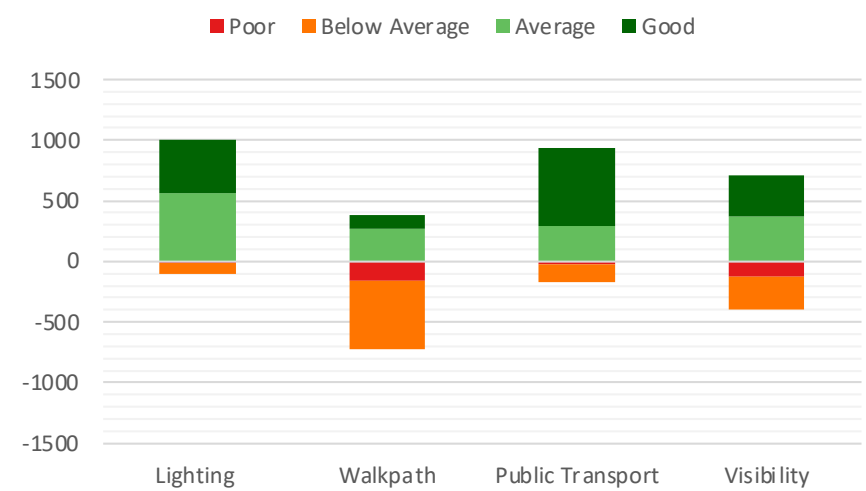
Tondiarpet: Average Parameter Graph



PARAMETER RATING

Graph top left
The graph shows average rating for each parameter on a scale of three.

Tondiarpet: Pin Distribution Graph



PIN DISTRIBUTION

Graph bottom left
The graph shows parameter wise pin distribution on a scale of three.

ACCESSIBILITY MAPPING FINDINGS

Accessibility maps and associated findings

ACCESS FOR ALL

Everyone is a pedestrian. Walking is a basic and common mode of transport in all societies across the world. Virtually, every trip begins and ends with walking. Walking comprises the sole means of travel on some journeys, whether a long trip or running an errand. Therefore, footpaths are a fundamental form of urban infrastructure that facilitate walking, socializing and interacting with the public domain. In order to have a safe, barrier-free and successful trip, the street systems must be accessible to all pedestrians and especially to vulnerable users - women, children, elderly and people with disabilities using mobility devices such as wheelchairs and walkers, those who have vision, speech, hearing impairments or those with intellectual disabilities. The ease with which pedestrians can access and use streets influences their everyday life choices. A good street design which enables the user to seamlessly complete a trip from doorstep of origin to the doorstep of the destination is critical.

In the absence of accessible footpaths, pedestrians are forced to share the street with motorists which enhances the chances of pedestrian injuries, disablement and fatalities. Absence or poor condition of footpaths also drastically restrict access to public spaces, services and amenities especially for women. It has a direct impact on their mobility patterns. It influences their ability to move around, connect to wider transport networks and access the opportunities present in the city. Therefore, footpaths impact the quality of their daily life and even their life choices.

Because streets are an important component of public space, designing them with people in mind is the key. Therefore, it is important to go beyond just having a footpath, it means that careful attention must be given to providing a comfortable, safe, engaging experience to all users of the street.

ACCESSIBILITY MAPPING OF TONDIARPET

To map accessibility in public spaces of Tondiarpet, especially for women users, Safetipin Nite application was used to map the entire area twice - once in the day and once again in the night. An additional set of accessibility parameters apart from the standard safety parameters mentioned in the previous chapter were applied for better understanding of the status of accessibility in the area, especially for the vulnerable groups. Day time images revealed the actual condition and usage of the footpaths when in maximum use during the working hours. Night time images revealed the street lighting condition clearly in the evening hours apart from footpath condition and usage. Both sets of day and night images were then put together to analyse the ease of accessing the footpaths in the area, considering walking as a primary way of access.

To map this, two sets of factors were considered to assess ease of walking on footpaths. First - to map the present condition of the footpaths. Second - to see if the footpaths are clear of any barriers or not. Based on this, the accessibility parameters considered are as given below. Composite maps are generated for each parameter with geo-tagged information which helps to assess the nature of the issues related to accessibility and locate the same for physical intervention at site.

Accessibility parameters used for mapping:

Footpath Conditions:

Non-existent footpath

Broken, unpaved, dis-continuous footpath

Availability of curb ramps, median breaks and zebra crossings

Footpath Barriers:

Garbage, construction debris/material dumped on footpaths

Vehicles parked on footpaths

House and shop extensions onto footpaths

Temporary stalls and vendors obstructing footpaths

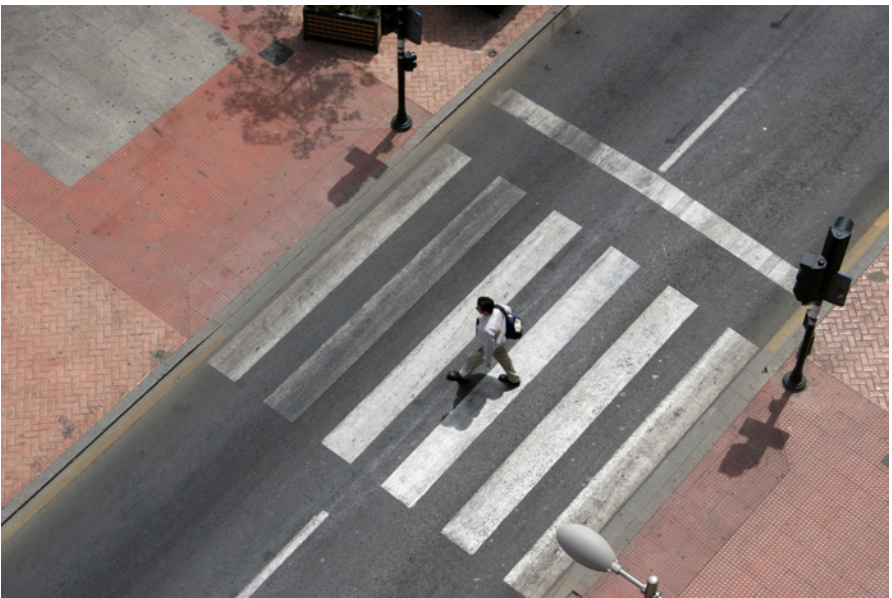
Light, electric, signage and advert poles obstructing footpath



CURB RAMPS

Image on left

The image shows correctly designed, located and executed curb ramps.



ZEBRA CROSSING

Image on left

The image shows correctly designed, located and executed pedestrian crossings.



MEDIAN BREAK

Image on left

The image shows correctly designed, located and executed median breaks.

MAPPING OF FOOTPATH CONDITIONS

Safetipin identified three types of footpath conditions which mar pedestrian walking experience and cause distress among pedestrians while navigating the streets, especially the vulnerable users. These conditions are namely, one, having no dedicated walking space on the streets, two, having no built footpath, and three, having broken, partly unpaved, discontinuous walking surface on footpaths. We mapped all of the above conditions wherever found in the area and geo-located the information on GIS layers. The composite map below represents all the non-functional footpaths as found on the streets and highlights the intensity of the issue in a large area in the middle of Tondiarpet.



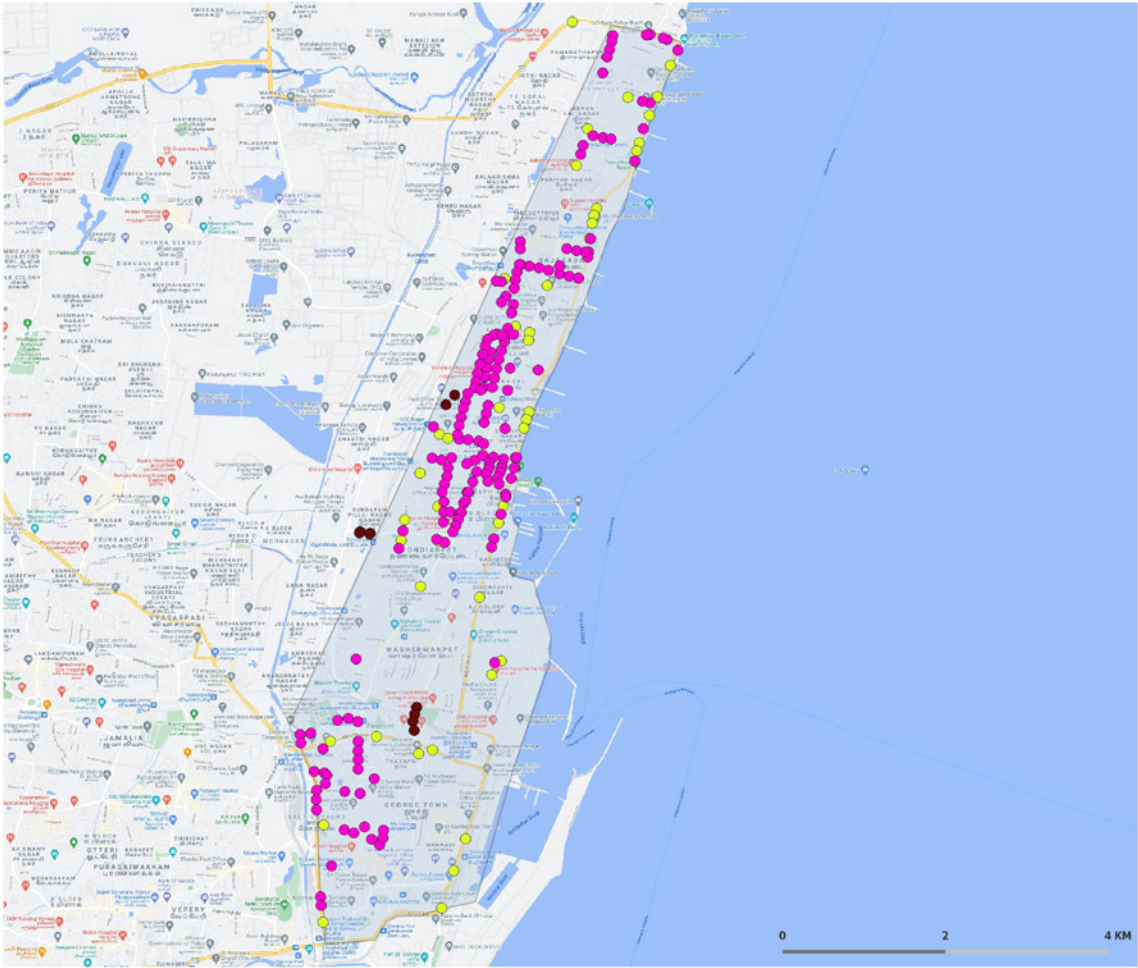
GEO-LOCATED IMAGES

Left to right

The geo-tagged photos show non-functional footpaths

Tondiarpet: Footpath Conditions

- Legend
- No Space for Footpath (10)
 - No Built Footpath (171)
 - Broken Footpath (44)
 - Tondiarpet
- Base: Google Map



MAPPING OF ACCESSIBILITY ELEMENTS

Safetipin identified three types of physical elements which assist pedestrians to have at grade walking experience, especially the vulnerable users. These are namely curb ramps, median breaks and zebra crossings. We mapped all of the above wherever found in the area and geo-located the information on GIS layers. The composite map below represents all the accessibility elements found on the streets and highlights the intensity of the issue of having inadequate curb ramps, median cuts and pedestrian ramps for a seamless, at grade walking experience. One pedestrian underpass and two foot-over bridges were located. Zebra crossings were found only on few locations and mainly on a north-south arterial road.



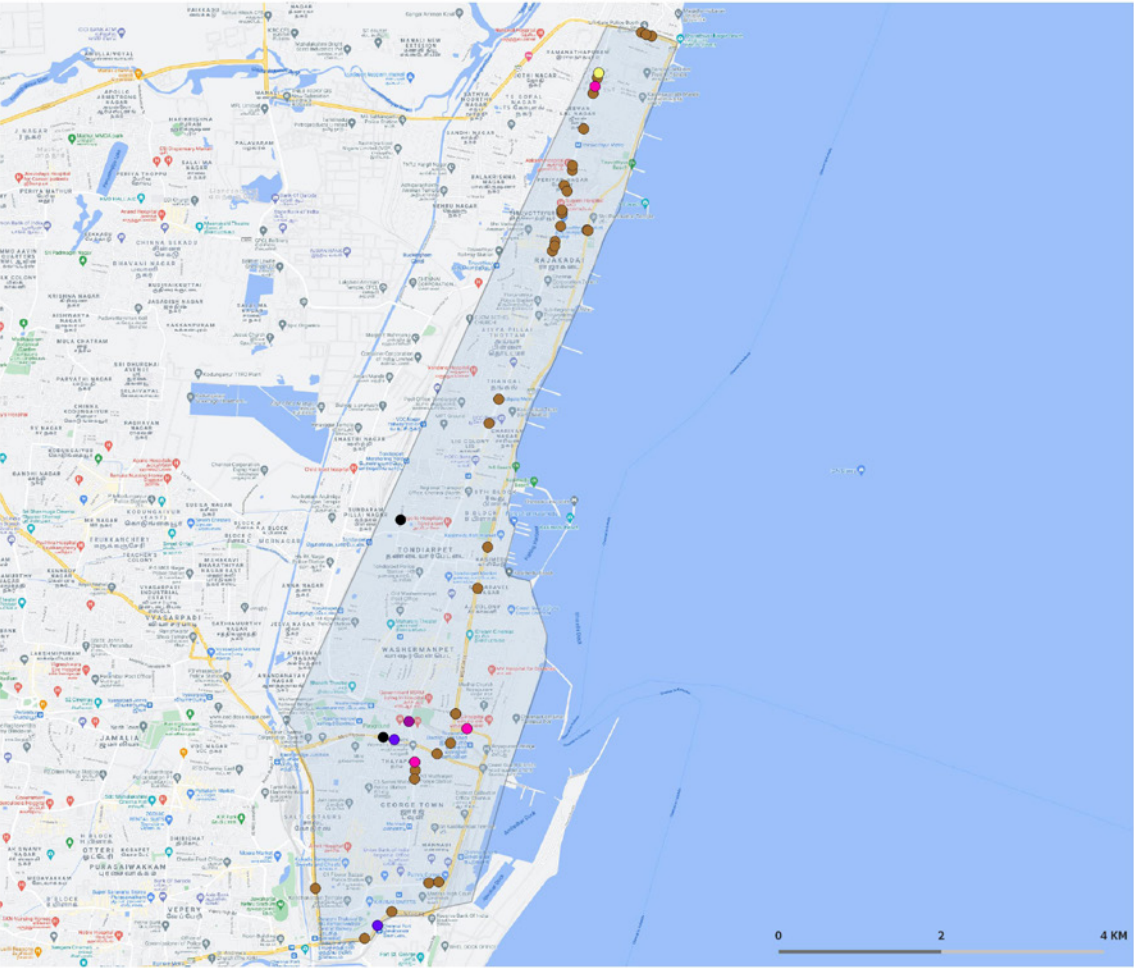
GEO-LOCATED IMAGES

Left to right

The geo-tagged photos show accessibility elements on streets

Tondiarpet: Accessibility Elements

- Legend
- Curb Ramp (3)
 - Median Cut (2)
 - Zebra Crossing (36)
 - Pedestrian Ramp (3)
 - Footover Bridge (2)
 - Pedestrian Underpass (1)
 - Tondiarpet
- Base: Google Map



MAPPING OF FOOTPATH OBSTRUCTIONS

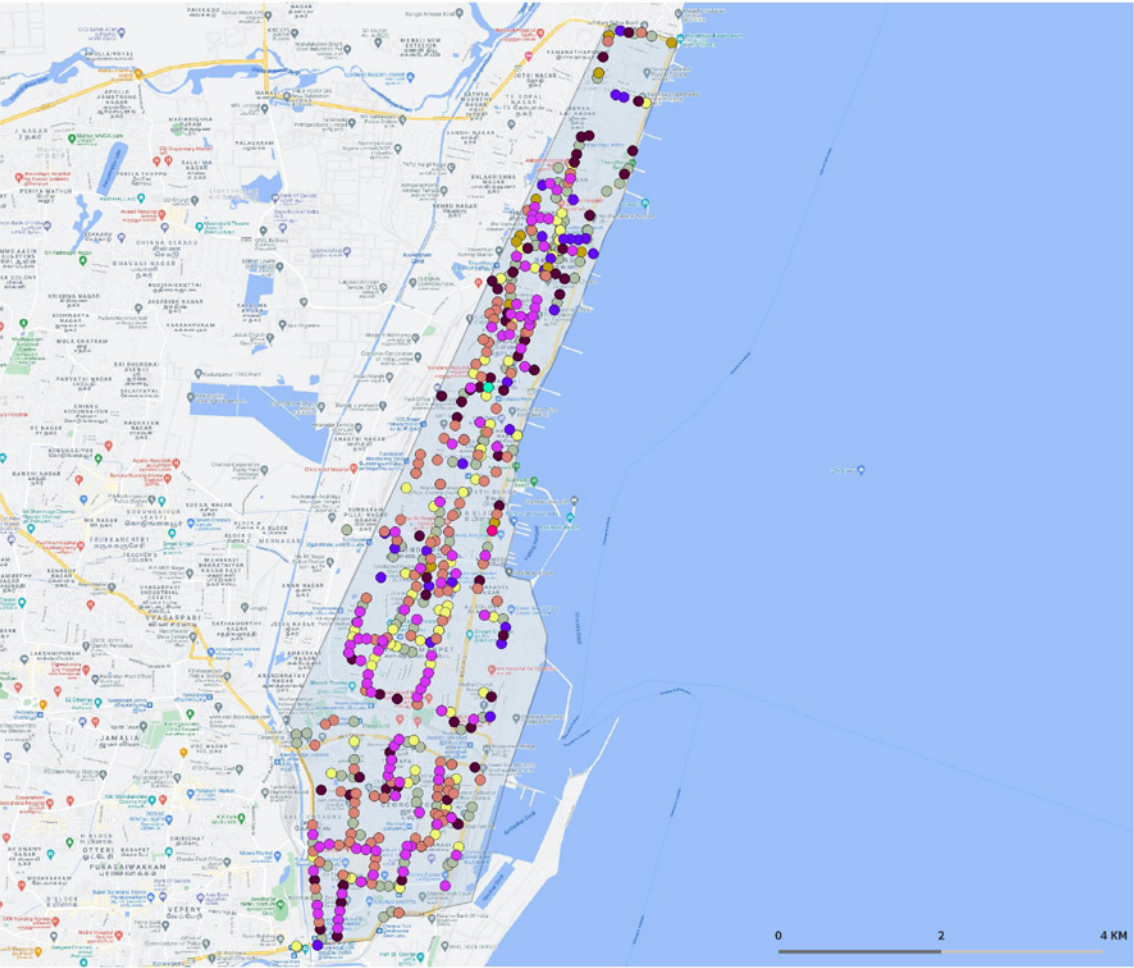
Safetipin identified five types of physical barriers disrupting clear walking space for pedestrians, especially the vulnerable users. These are namely parked vehicles, shop/house extensions, temporary stalls/ vendors, electricity/signage poles, and lastly dumped garbage/construction material. We mapped all of the above when found on footpaths and geo-located the information on GIS layers. The composite map below represents all barriers types found on footpaths and highlights the intensity of the issue. The total number of locations where these obstructions are found are indicated in the legend.



GEO-LOCATED IMAGES
Left to right
The geo-tagged photos show obstructions on footpaths

Tondiarpet: Footpath Obstructions

- Legend
- Shop Encroachment (92)
 - Repair Work (2)
 - Light Pole (1)
 - Electricity Pole (132)
 - Garbage Dump (77)
 - Construction Debris (16)
 - Construction Material (42)
 - Signage Pole (125)
 - Stalls/Vendors (208)
 - Tondiarpet
- Base: Google Map



MAPPING OF FOOTPATH OBSTRUCTIONS - PARKING

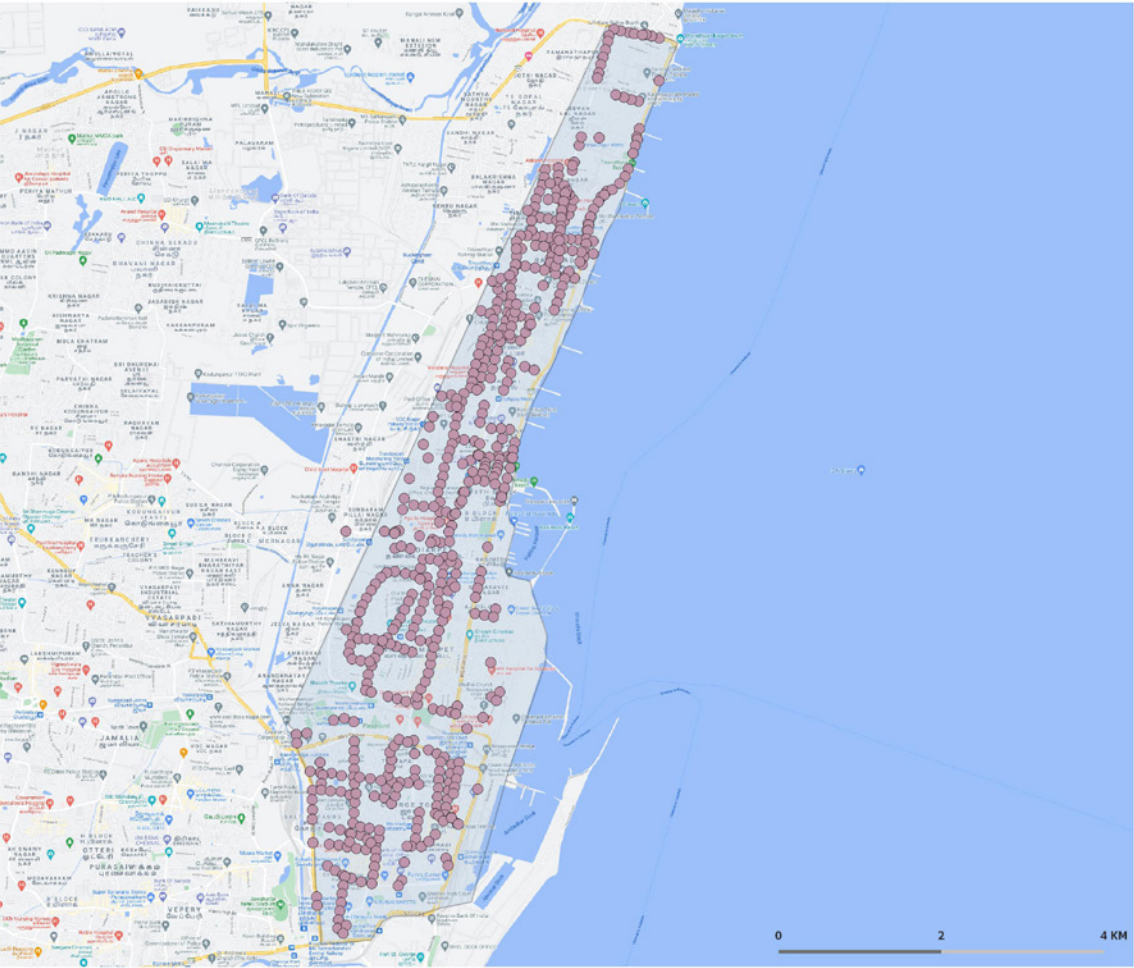
Safetipin identified 544 locations within Tondiarpet where vehicles are parked on footpaths, disrupting clear walking space for pedestrians, especially the vulnerable users. These are mainly two wheelers, four wheelers, small delivery vehicles, auto rickshaws, carts and cycles. We mapped these when found parked on footpaths and geo-located the information on GIS layers. The composite map below represents all locations with parked vehicles and highlights the intensity of the issue. It is found that majority of the footpaths were either occupied by obstructions or did not have a good, at grade, walking surface.



GEO-LOCATED IMAGES
Left to right
The geo-tagged photos show parked vehicles on footpaths

Tondiarpet: Footpath Obstructions

- Legend
- Parked Vehicles (544)
 - Tondiarpet
- Base: Google Map



DATA ANALYSIS

Data representation and correlation maps

To further analyse the data layers generated by Safetipin and to gain a deeper understanding of the spatial issues related to safety and accessibility, some data layers were overlaid onto other data layers. This helped in understanding their relationship with each other and how they interacted with space. Such understanding could be applied to inform decision making for spatial interventions. The following correlation maps explore this understanding at a spatial level. Safetipin also looked at representation of data layers by grouping and segregating them. For example the Safety Score map below illustrates Safety Score at each ward level and can be used by the development authorities to prioritise on the wards which appear light green on the map and have scored above average on safety and accessibility.

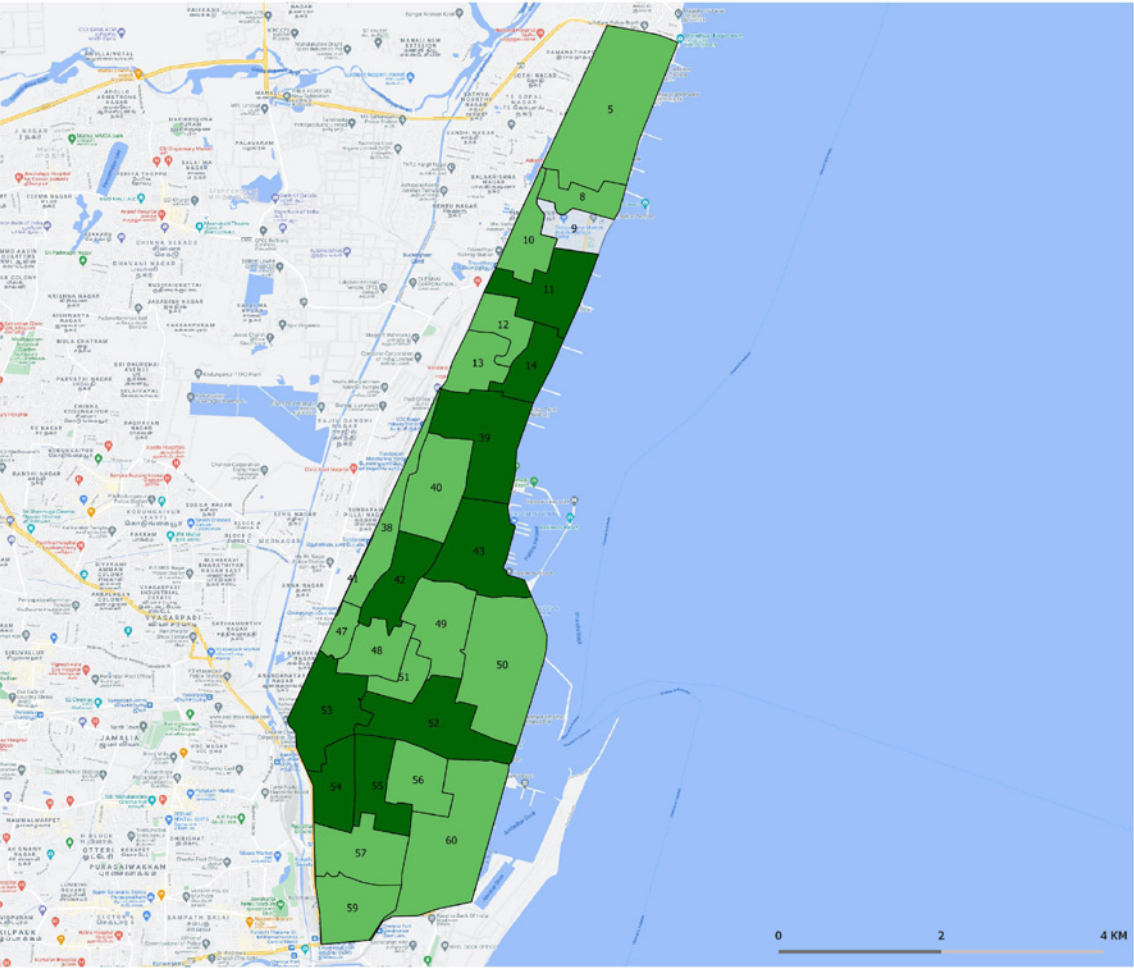
A set of recommendations on the actionable parameters for immediate improvement which will positively impact the wards' safety scores is included in the next chapter.

Ward-wise Safety Score map below showing good and above average safety score in various wards.

Chennai: Ward Wise Safety Score

Legend

- Safety Score
- Poor
- Below Average
- Average
- Above Average
- Good
- Chennai
- Base: Google Map



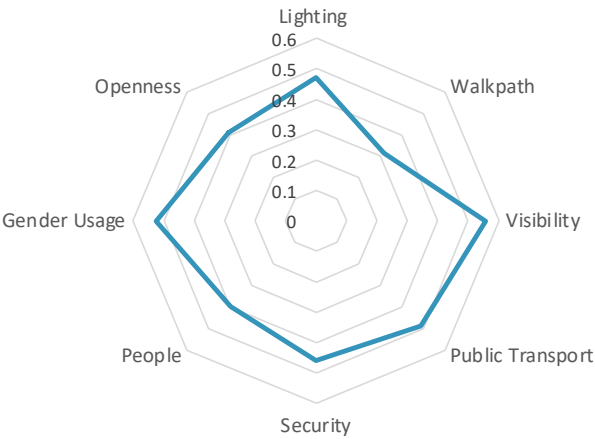
KEY PARAMETERS IMPACTING SAFETY PERCEPTIONS

The graph below indicates the correlation drawn between each of the eight parameters with respect to feeling, the ninth parameter. In case of Tondiarpet, user generated 'My Safetipin' audits show Visibility, Gender Usage and Public Transport to be rated the highest when correlated with feeling. This clearly shows the strong relationship of these parameters to women's perception of safety. Lighting and Security have been rated high as well. Walkpath however has been rated lower. The reason for this could be being used to walking on shared streets with transport and others activities happening on streets in an organically grown area like Tondiarpet. This analysis helps in identifying the key actionable parameters on which physical intervention is possible.

CORRELATION GRAPH

Graph on right

The graph shows every parameter correlation with the Feeling parameter.



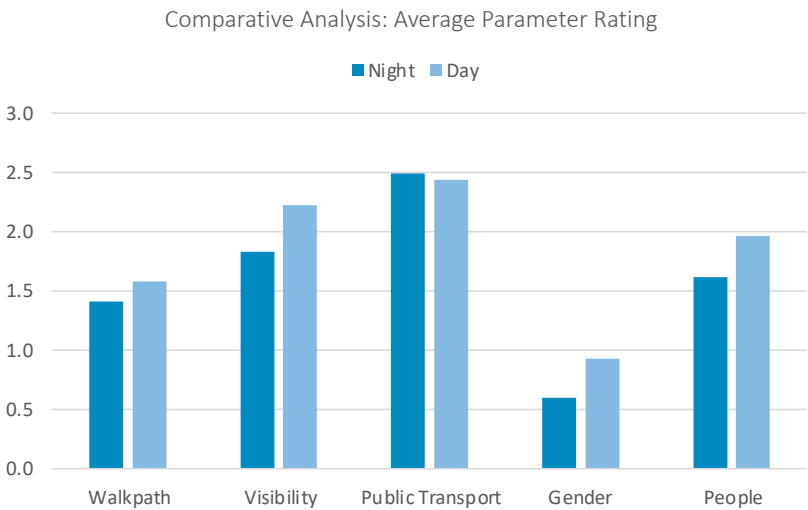
COMPARING DAY AND NIGHT AUDITS

Day and night audits generated using Safetipin Nite were compared for five parameters - Walkpath, Visibility, Public Transport, Gender and People. Lighting was not considered as street lights are not operational in the day. It is seen that Walkpath and Visibility drops at night along with Gender and People but Public Transport is slightly over day ratings. It is not very clear why public transport ratings are higher.

COMPARATIVE RATING

Graph on right

The graph shows average rating of key parameters for day and night audits



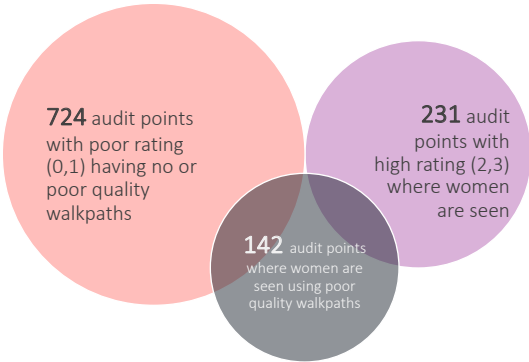
CORRELATING GENDER USAGE WITH WALKPATH

Safetipin overlaid 'presence of gender in public spaces' data on 'walkpath' data, generated by Safetipin's visual mapping analysis. This is done to geographically locate the areas where women and others are seen using walkpaths with various obstructions. The heat map below clearly illustrates this correlation. The dark purple areas are where walkpath is good and the light pink areas are where walkpath is non-existent or of poor quality. The blue dots are the audit points showing presence of gender. There are all together 142 audit points identified in Tondiarpet where women can be seen accessing the streets where the walkpath condition is poor. The diagram below represents this data quantitatively. Each geo-tagged audit point is supported by two or more images. Such photographic evidence could be used for deeper investigation into the area's walking infrastructure and improve accessibility in public spaces. A good walking environment will improve women's mobility and access to resources.

CORRELATION DIAGRAM

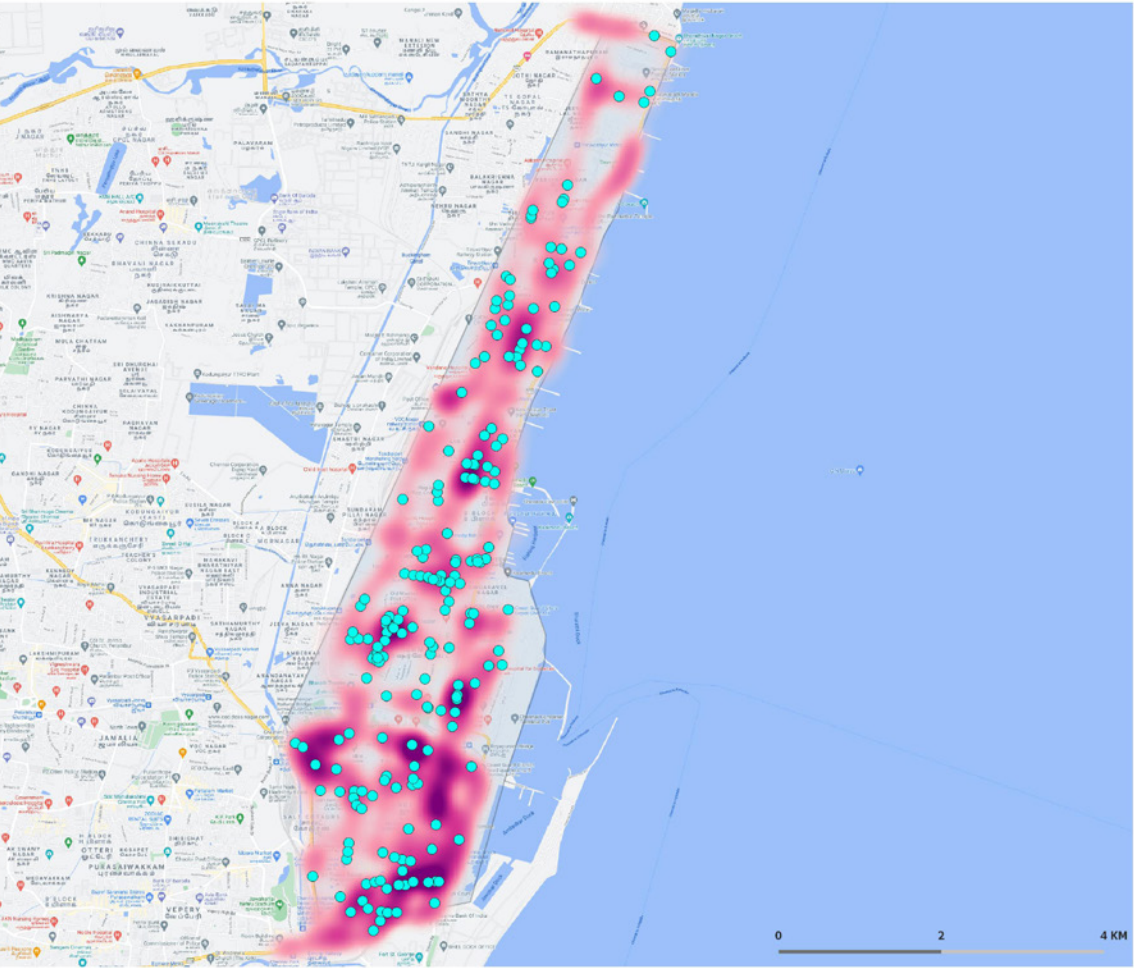
Diagram on right

The diagram represents correlation between gender usage and walkpath



Tondiarpet: Heat map showing Gender Usage and Footpaths

- Legend
- High Gender Usage
 - Walkpath
 - None
 - Poor
 - Fair
 - Good
 - Tondiarpet
 - Base: Google Map



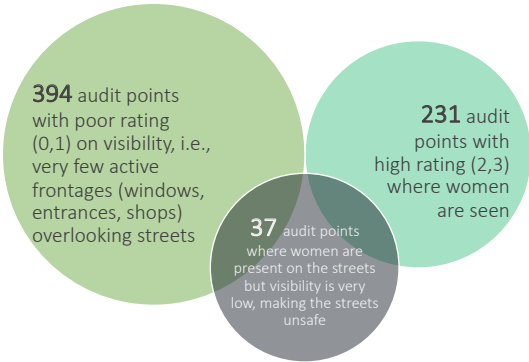
CORRELATING GENDER USAGE WITH VISIBILITY

Safetipin overlaid 'presence of gender in public spaces' data on 'visibility' data, generated by Safetipin's visual mapping analysis. This is done to geographically locate the areas where women and others are seen using public spaces which have very low visibility. The heat map below clearly illustrates this correlation. The dark green areas are where visibility or 'eyes on the street' is good and the light green areas are where there is no visibility. The blue dots are the audit points showing presence of gender. There are all together 37 audit points identified in Tondiarpet where women can be seen accessing public spaces after dark with no eyes on the streets. The diagram below represents this data quantitatively. Each geo-tagged audit point is supported by two or more images. Such photographic evidence could be used for further analysis of the area's safety aspects. Active and watched over public places encourage and enable women's participation in city life and enhance women's perception of safety.

CORRELATION DIAGRAM

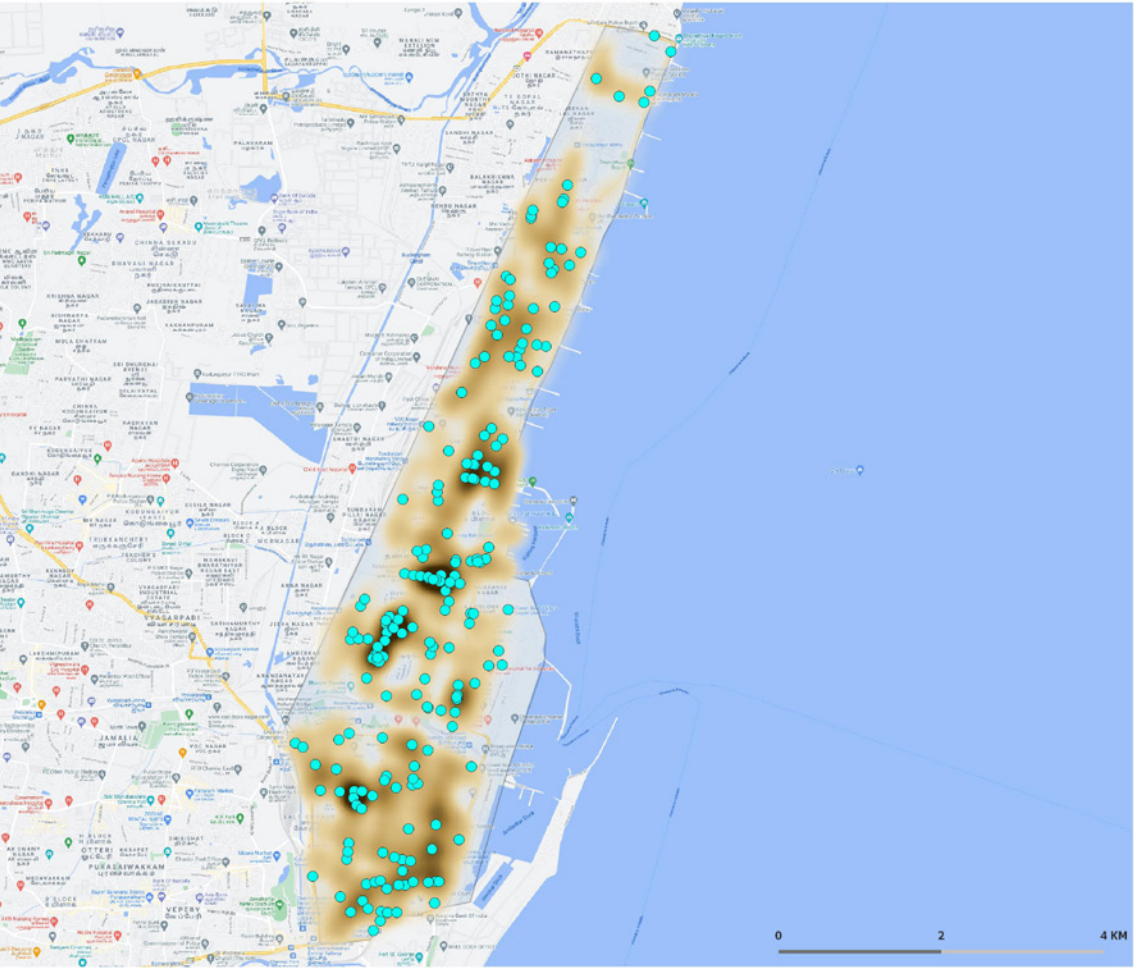
Diagram on right

The diagram represents correlation between gender usage and visibility



Tondiarpet: Heat map showing Gender Usage and Visibility

- Legend
- High Gender Usage
 - Visibility
 - No Eyes
 - Few Eyes
 - More Eyes
 - Highly Visible
 - Tondiarpet
 - Base: Google Map



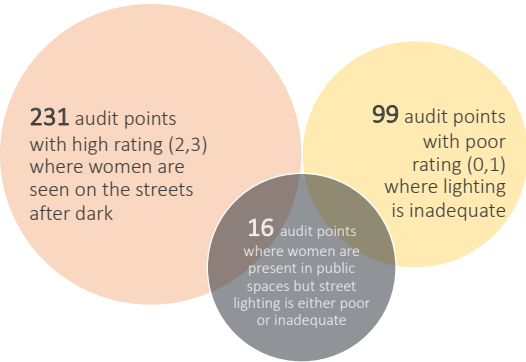
CORRELATING GENDER USAGE WITH STREET LIGHTING

Safetipin overlaid 'presence of gender in public spaces' data on 'street lighting' data, generated by Safetipin's visual mapping analysis. This is done to geographically locate the areas where women and others are seen using public spaces which have very low street lighting. The heat map below clearly illustrates this correlation. The dark brown areas are where street lighting is good and the light yellow areas are where street lighting is non-functional or inadequate. The blue dots are the audit points showing presence of gender. There are all together 16 audit points identified in Tondiarpet where women can be seen accessing public spaces in the dark. The diagram below represents this data quantitatively. Each geo-tagged audit point is supported by two or more images. Such photographic evidence could be used for deeper investigation into the area's street lighting infrastructure and improve safe access to public spaces. Women's access to public spaces would certainly improve if street lighting is prioritise.

CORRELATION DIAGRAM

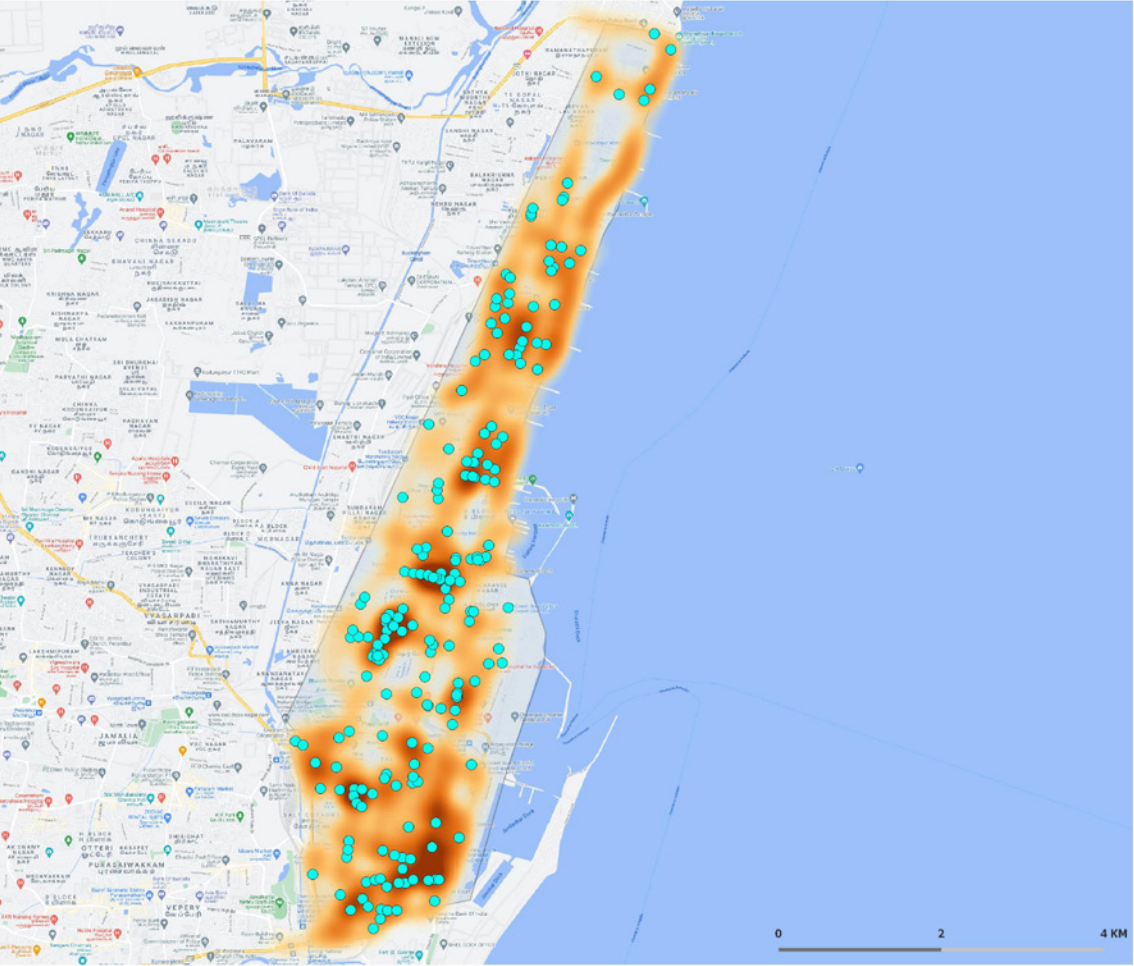
Diagram on right

The diagram represents correlation between gender usage and street lighting



Tondiarpet: Heat map showing Gender Usage and Street Lighting

- Legend
- High Gender Usage
 - Lighting
 - Poor Light
 - Some Light
 - Enough Light
 - Bright Light
 - Tondiarpet
 - Base: Google Map



CORRELATING PUBLIC TRANSPORT WITH PUBLIC SPACES

Safetipin overlaid public transport data on a GIS layer of public places (markets, transport hubs, schools, colleges, hostels) received from the city authorities. This helped in geographically locating the areas around public places where public transport stands/stops are available or unavailable. The heat map below clearly illustrates this correlation. The dark green areas are where transport stands are within reach and the light yellow areas are where they are out of reach. This GIS layer is provided as a CSV file and could be integrated with the city's existing GIS platform for an up-gradation program to make public transport available within 2-5 mins of walking distance from the schools which fall in the light yellow zone.

GEO-LOCATED IMAGES

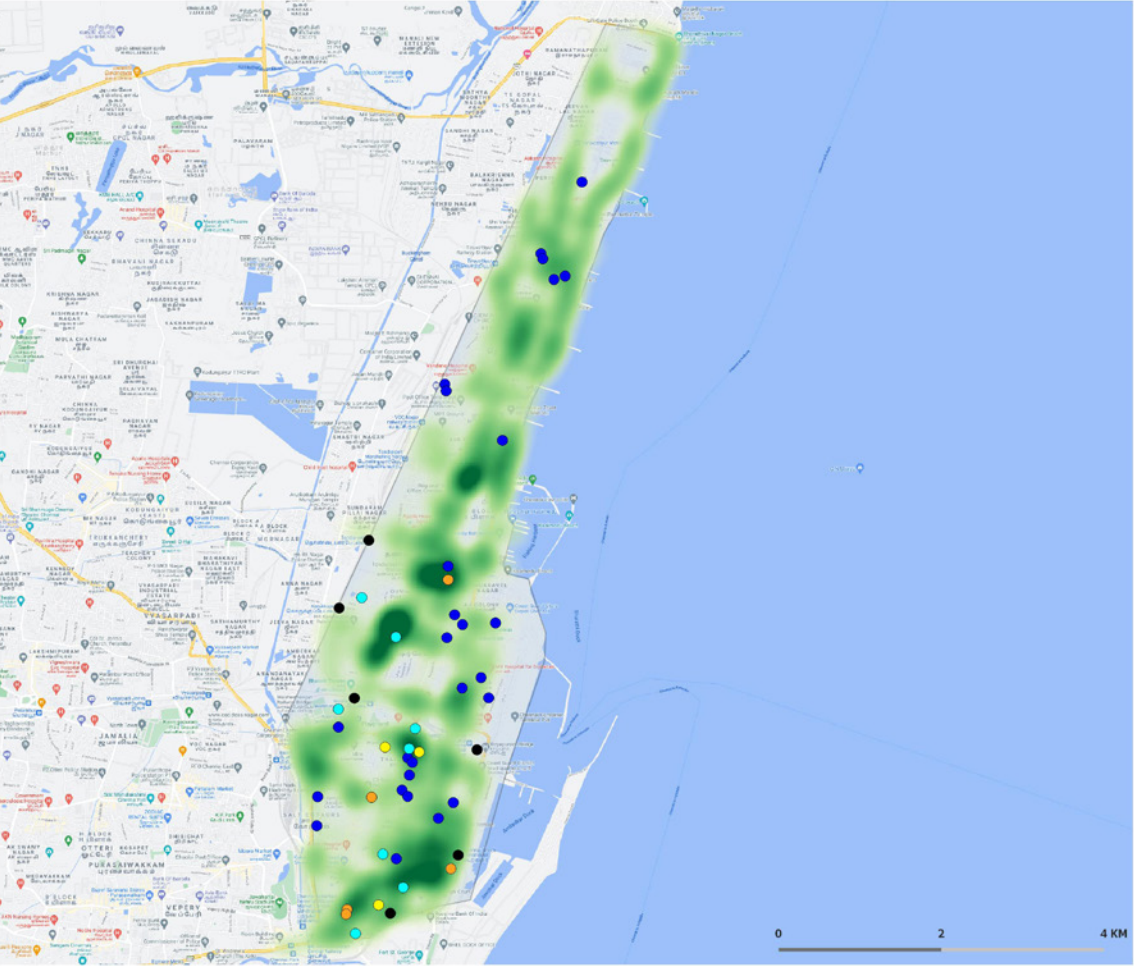
Left to right

The geo-tagged photos show public transport stands/stops.



Tondiarpet: Heat map showing Public Transport around Public Spaces

- Legend
- School (27)
 - College (8)
 - Hostel (3)
 - Market (5)
 - Railway Station (6)
 - Transport
 - Unavailable
 - Distant
 - Nearby
 - Very Close
 - Tondiarpet
 - Base: Google Map



CORRELATING SAFETY SCORE WITH SAFETY POLES

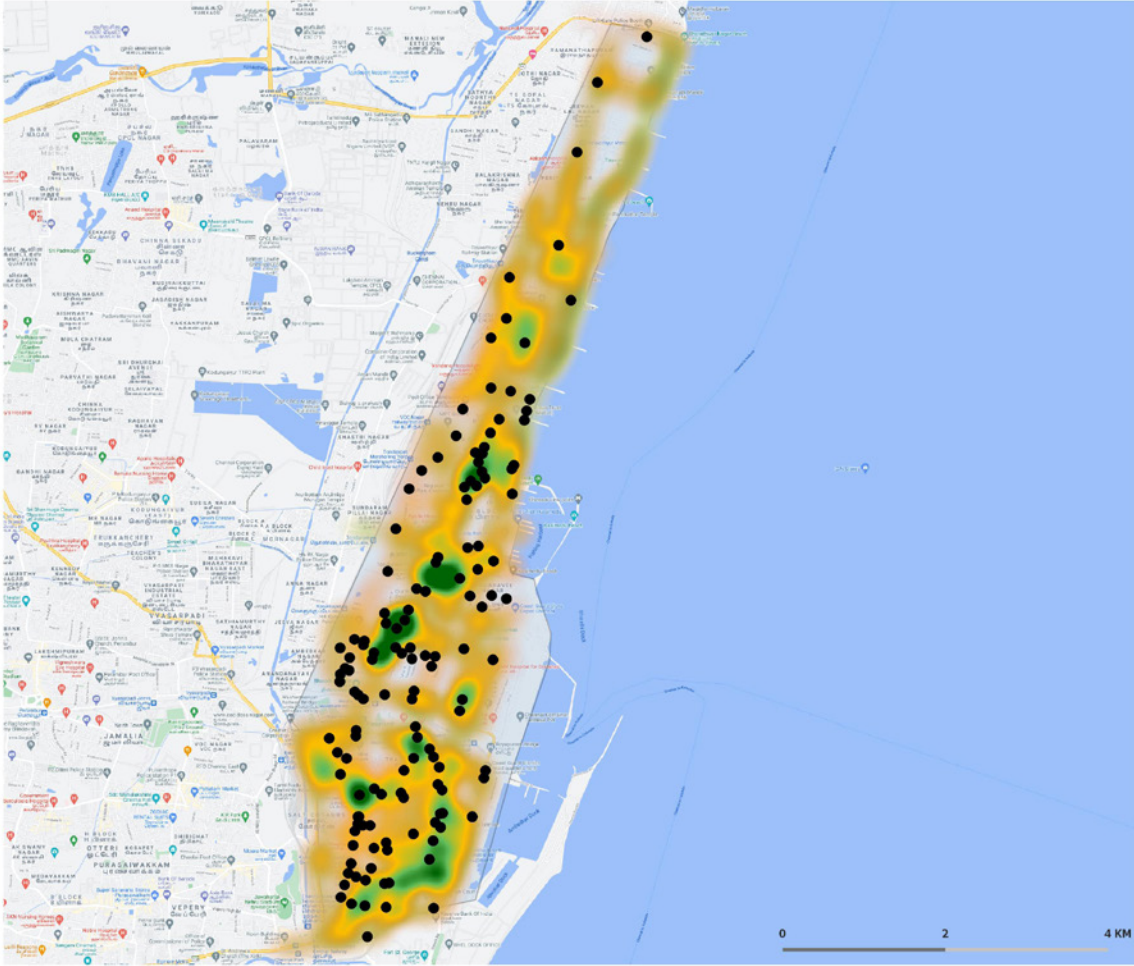
Safety Score and Safety Pole

Proposed Safety Pole locations were shared with Safetipin by the project team at World Bank. We overlaid the locations on the Safety Score heat map to see if they are positioned over areas with good or poor safety score. The map below illustrates this relationship.

While the safety pole program has been put on hold by the authorities, the analysis provides an opportunity to better understand where such intervention could be most effective.

Tondiarpet:
Heat map showing
Safety Score around
Safety Poles

- Legend
- Safety Poles
 - Safety Score
 - Poor
 - Below Average
 - Average
 - Above Average
 - Good
 - Tondiarpet
- Base: Google Map



CORRELATING GENDER USAGE WITH SAFETY POLES

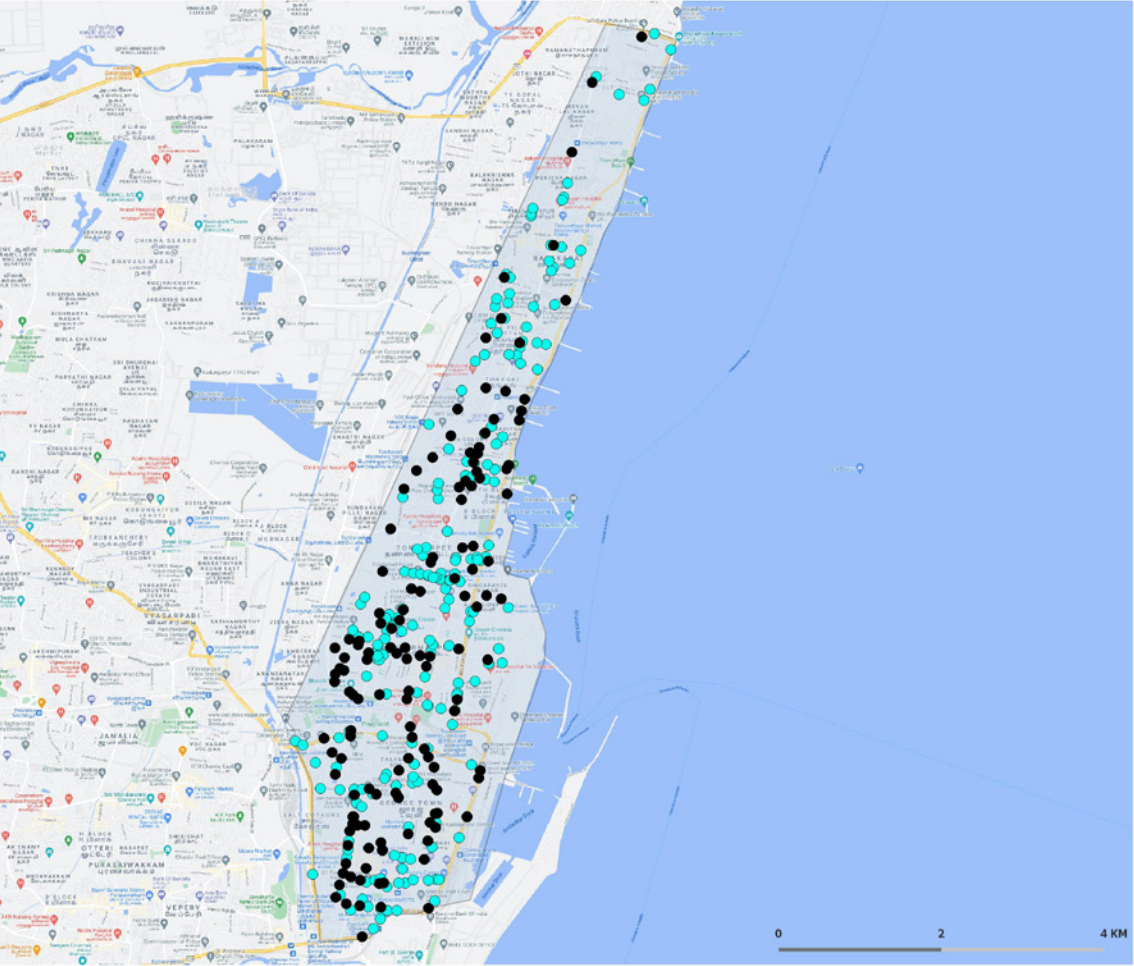
Gender Usage and Safety Pole

We also overlaid the safety pole locations on 'presence of gender in public spaces' parameter to see if they are positioned over areas where high presence of gender were found during mapping. The map below illustrates this relationship.

While the safety pole program has been put on hold by the authorities, the analysis provides an opportunity to better understand where such intervention could be most effective.

Tondiarpet:
Correlation between
Safety Poles and
Gender Usage

- Legend
- Safety Poles
 - High Gender Usage
 - Tondiarpet
- Base: Google Map



RECOMMENDATIONS

Action points and planned interventions

KEY ACTION: MAKE PUBLIC SPACES WALKABLE

Walkable cities have places that are easily and safely navigable on foot. Walking has the capacity to promote equality and reduce social exclusion. As a free means of transport, walking can provide access to a range of resources and opportunities. Persons with disabilities (people walking with walking aid), people assisting others (pushing wheelchairs and prams) and people having reduced walking speed (children and elderly) - must have equal access to the walking facility. This implies that people on foot are given consideration first in street design by keeping the city's most vulnerable user in mind. This is the core principle of universal access: every street must be accessible by people of any age and ability.

Tondiarpet has a great potential to become a pedestrian priority area with good local level pedestrian connections by improving the quality of footpaths, walking environments and links to access public transport.

As discussed before, the analysed data highlights the state of physical infrastructure and social participation in public space. Hence, in order to make streets and public spaces more safe and accessible, the following parameters need immediate attention. Condition of street lighting, quality of footpath and access to public transport, are the physical infrastructure parameters and visibility, is the social usage parameter which need improvement. Specific maps for each parameter with data points having lower ratings are given below. Analysed maps are also included with specific recommendations on wards, bus stops, markets and educational institutions. A detailed recommendation on walkability is given as well which looks at immediate upgradation of the walking environment of Tondiarpet.

LEARNING FROM OTHER CITIES

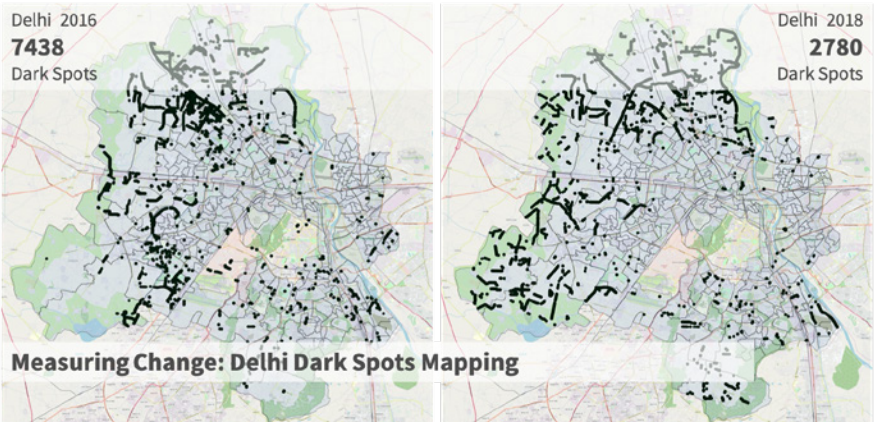
Safetipin's mapping in other cities like Delhi, Gurgaon, Gwalior have yielded positive results in identifying and improving certain parameters for enhancing safety and accessibility in public spaces.

In Delhi in 2014, after the first round of mapping, 7438 darkspots were identified in the city. The authorities made dedicated efforts in making the city well lit at night to improve safety and usage of public space. In 2018, another round of audits were done to map the city's improved condition and to identify more dark spots.

Similarly, in Gurgaon in 2016, mapping was done to improve the quality of street lighting in order to make people feel safer on the streets after dark. The Gurgaon Municipal Corporation had asked to identify places where people were seen at night and where street lighting was inadequate. 630 nos. of such spots were located in the city where street lighting was improved for better visibility and safety at night.

In Gwalior in 2019, after Safetipin's mapping of the city, Gwalior Smart City launched 'Gwalior One City One App' to provide its citizens information on administrative as well as infrastructural status of the city. One section of the app was developed and named 'Veerangana' (meaning brave woman) that allows users to conduct safety reviews. This review is similar to Safetipin app's Safety Audit wherein one has to rate physical and social infrastructure of a place on the basis of its condition and status.

- Analysed data shows,
- 65% of Tondiarpet, has no or poor quality footpath
 - 15% of Tondiarpet has distant or unavailable transport stops/stands within 5-10 minutes of walking distance
 - 35% of Tondiarpet has no or poor visibility
 - 40% of Tondiarpet has deserted public spaces
- Women and children using public space in the evening (6-9pm)
- 20% of Tondiarpet, have presence of women & children in streets



DELHI AUDIT

Image on left

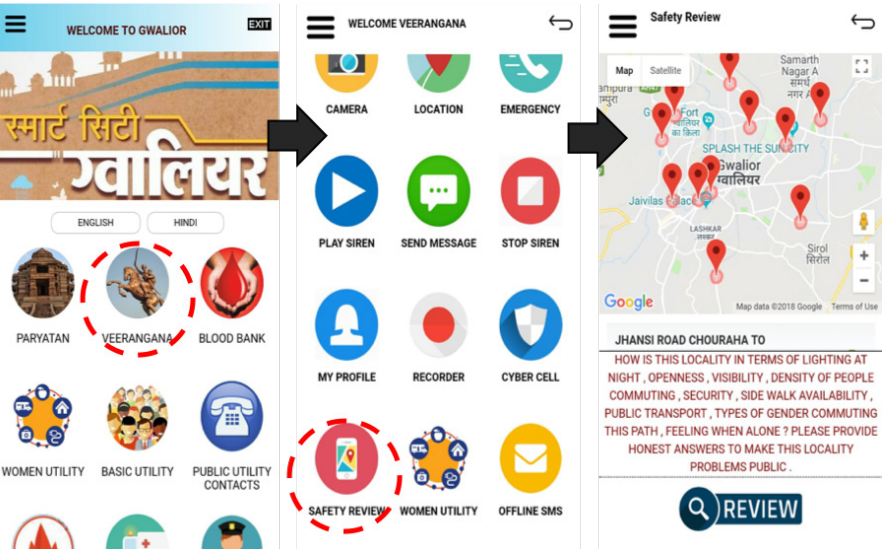
The image shows dark spot mapping of the city of Delhi



GURGAON AUDIT

Image on left

The image shows before and after photos of a stretch of road where lighting was improved



GWALIOR AUDIT

Image on left

The image shows Smart City Gwalior app with Safety Review feature

1. IMPROVE FOOTPATH CONDITION TO ENHANCE WALKABILITY

65% of the area is either without any footpath or has very poor footpath condition. Broken, discontinuous and blocked footpaths could be easily fixed to achieve ease of walking in the area which is important for pedestrians, especially women to feel safe while accessing the area for various daily activities. Safetipin's data (geo-located images) can be used to improve pedestrian access in Tondiarpet. The map below represents the locations where there are no dedicated walkpaths, no constructed footpaths and footpaths with obstructions. This information can be used for footpath up-gradation such as, construction, repair and removal of obstructions. More detailed recommendations on footpaths follows.



GEO-LOCATED IMAGES

Left to right

The geo-tagged photos show very narrow footpath.

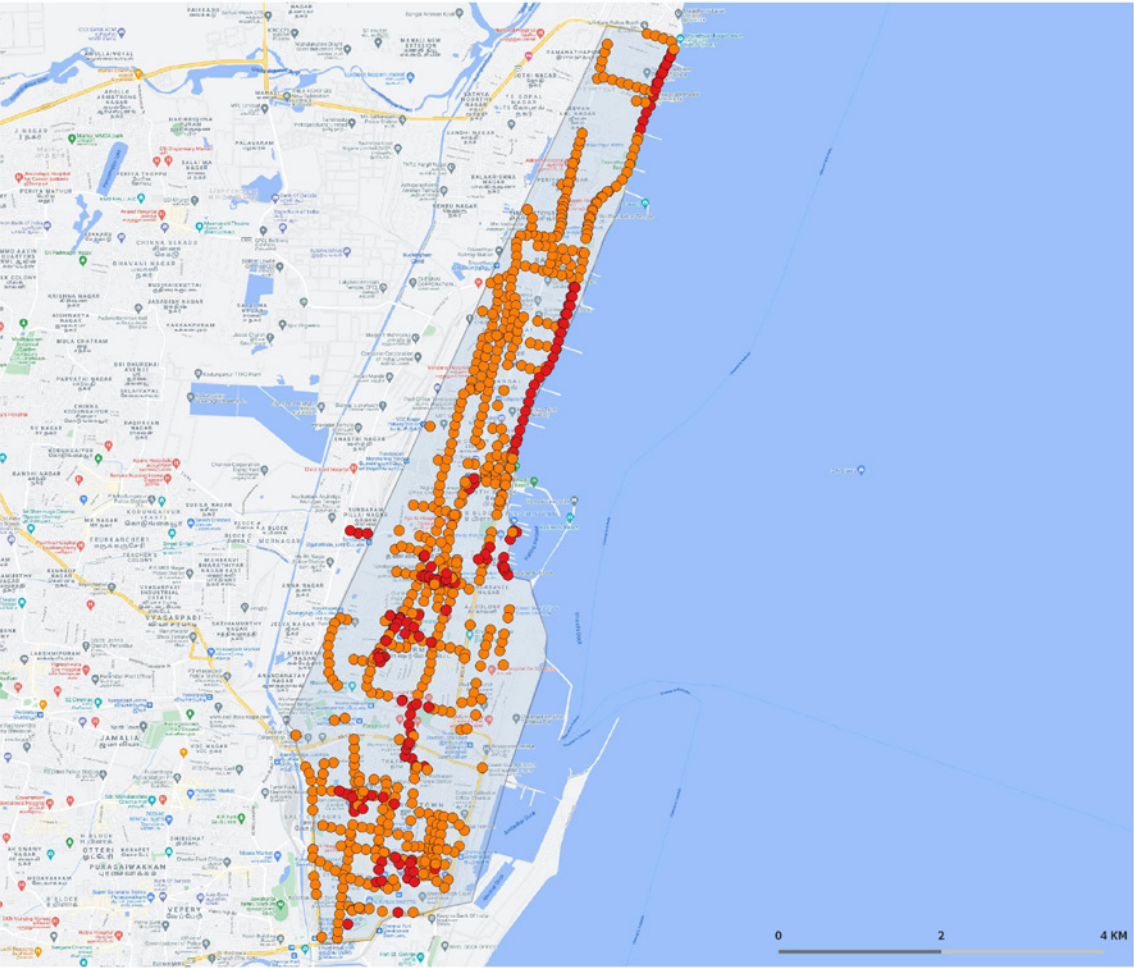
Tondiarpet: Poor Walkpath

Legend

Safety Audits

- None (158)
- Poor (566)
- Tondiarpet

Base: Google Map



2. MAKE STREETS VISIBLE OR ADD MORE 'EYES ON THE STREETS' TO THEM SAFER

35% of the area has either not any or poor visibility which means people present on the streets cannot be seen by other people living by or working on the streets - the concept of 'eyes on the streets'. Safetipin's data (geo-located images) can be used to improve 'eyes on the streets' - the key component in making cities safe and accessible. The map below represents the locations where visibility is low. This information could be used for enhancement of visibility by replacing opaque, high, boundary walls with fences, adding lights to transport stops, foot-overs, underpasses and allowing/organising street hawking. If public spaces are active it positively impacts women's perception of safety in the city.



GEO-LOCATED IMAGES

Left to right

The geo-tagged photos show good visibility.

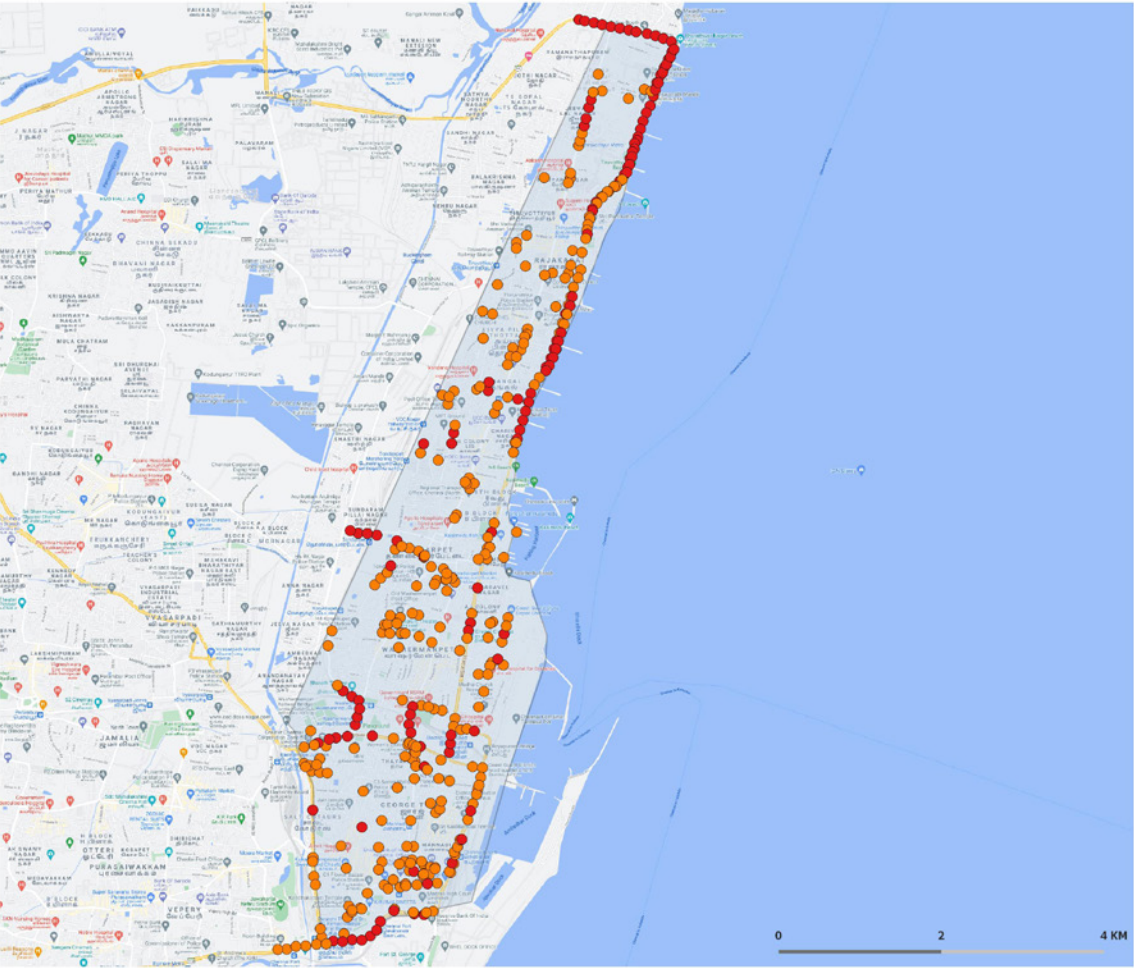
Tondiarpet: Poor Visibility

Legend

Safety Audits

- No Eyes (125)
- Few Eyes (269)
- Tondiarpet

Base: Google Map



3. MAKE PUBLIC TRANSPORT ACCESSIBLE TO ENHANCE REACH TO OPPORTUNITIES

Only 15% of the area has either distant or out of reach public transport. Adding more public transport stops/stands will improve access for all. Safe and reliable public transport is the critical link between women's access to education, employment and resources in the city. Safetipin's data (geo-located images) can be used to improve access to public transport Tondiarpet. The map below represents the locations where there are either no public transport stops available or the stops are distant (beyond 10 minutes walking distance). This information can be used for planning transport linkages such as, adding more bus stops, auto/rickhaw stands within 2-5 minutes walking distance on the identified locations.



GEO-LOCATED IMAGES

Left to right

The geo-tagged photos show public transport stands.

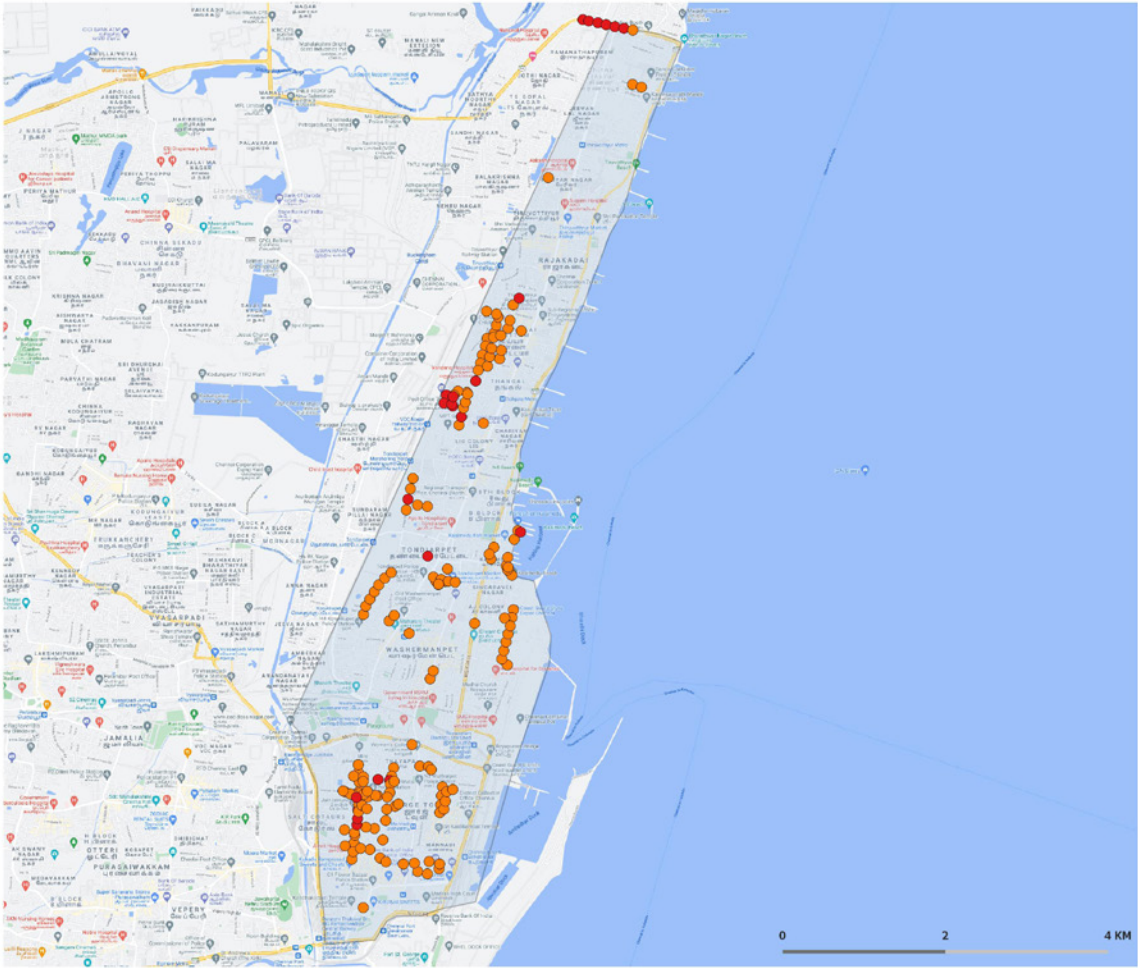
Tondiarpet: Poor Public Transport

Legend

Safety Audits

- Unavailable (26)
- Distant (146)
- Tondiarpet

Base: Google Map



4. MAKE STREET LIGHTING FULLY FUNCTIONAL TO IMPROVE SAFETY AND ACCESS

Only 09% of the area has inadequate street lighting and could be fixed immediately to achieve 100% street lighting in Tondiarpet. Street lighting is crucial for ensuring safe accessibility in public spaces and plays a vital role in people's, particularly women's perception of safety in streets and public places after dark. Safetipin's data (geo-located images) can be used to improve street lighting in the city. The map below represents the locations where there are either absolutely no lights, inadequate lights or non-functional street lights. This information can be used for the ease of planning for immediate up-gradation like, adding and fixing street lights in order to improve women's access to public spaces.



GEO-LOCATED IMAGES

Left to right

The geo-tagged photos show low street lighting.

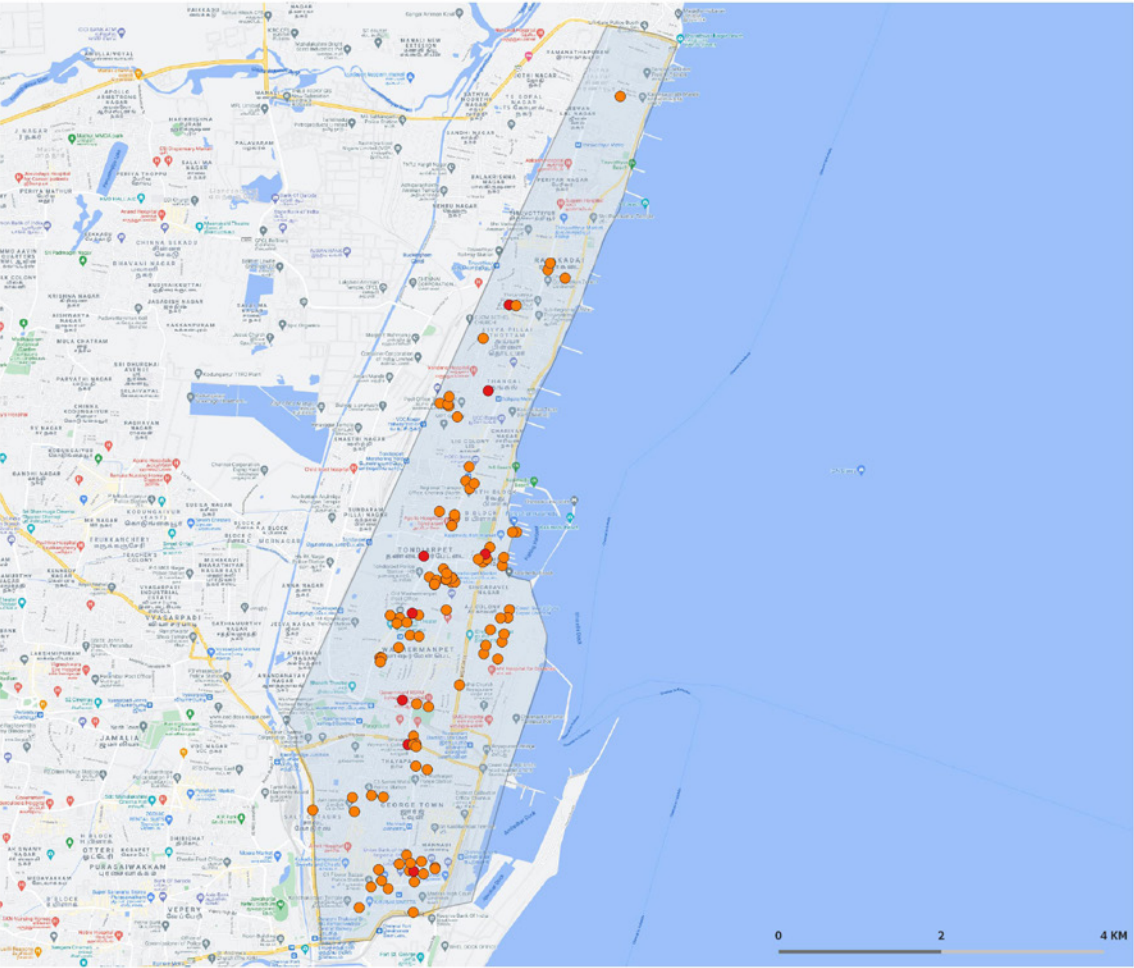
Tondiarpet: Poor Lighting

Legend

Safety Audits

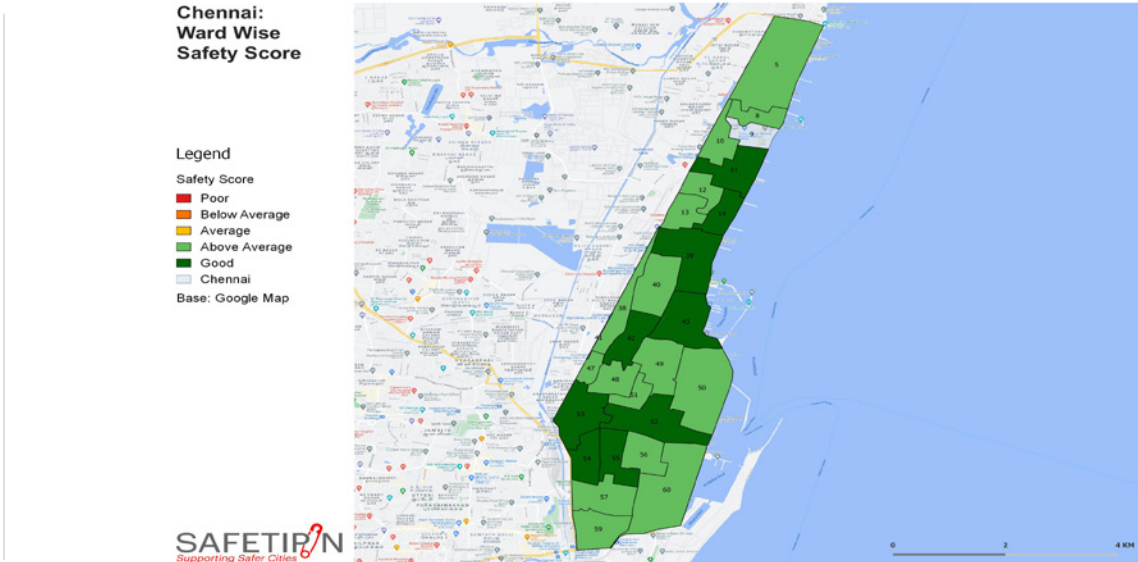
- Poor Light (8)
- Some Light (91)
- Tondiarpet

Base: Google Map



5. IMPROVE FOOTPATH, VISIBILITY, PUBLIC TRANSPORT FOR WARDS WITH AVERAGE RATING

Lighting, Walkpath, Public Transport and Visibility are the parameters which address some of the key issues of safety and accessibility. Conditions of Walkpath and Public Transport directly affects accessibility and mobility in the area, especially for women. Whereas, Lighting and Visibility affects actual safety as well as perception of safety on the streets by the availability of street lighting after dark and by the presence of ‘eyes on the streets.’ These parameters along with Gender Diversity have the strongest correlation with the overall safety score. If physical infrastructure parameters - Lighting, Walkpath, Public Transport and Visibility could be improved with strategic interventions at neighbourhood and city level - collectively these will have a positive impact on Gender Diversity which is a parameter which measures social usage of space. Improving these parameters will impact safety and accessibility in 17 wards of Tondiarpet (light green as seen on the map below) which have scored 'above average' on safety rating. These ward numbers are 5, 8, 10, 12, 13, 38, 40, 41, 47, 48, 49, 50, 51, 56, 57, 59 and 60. A set of recommendations on the actionable parameter are given below for these wards to improve their overall Safety Score.



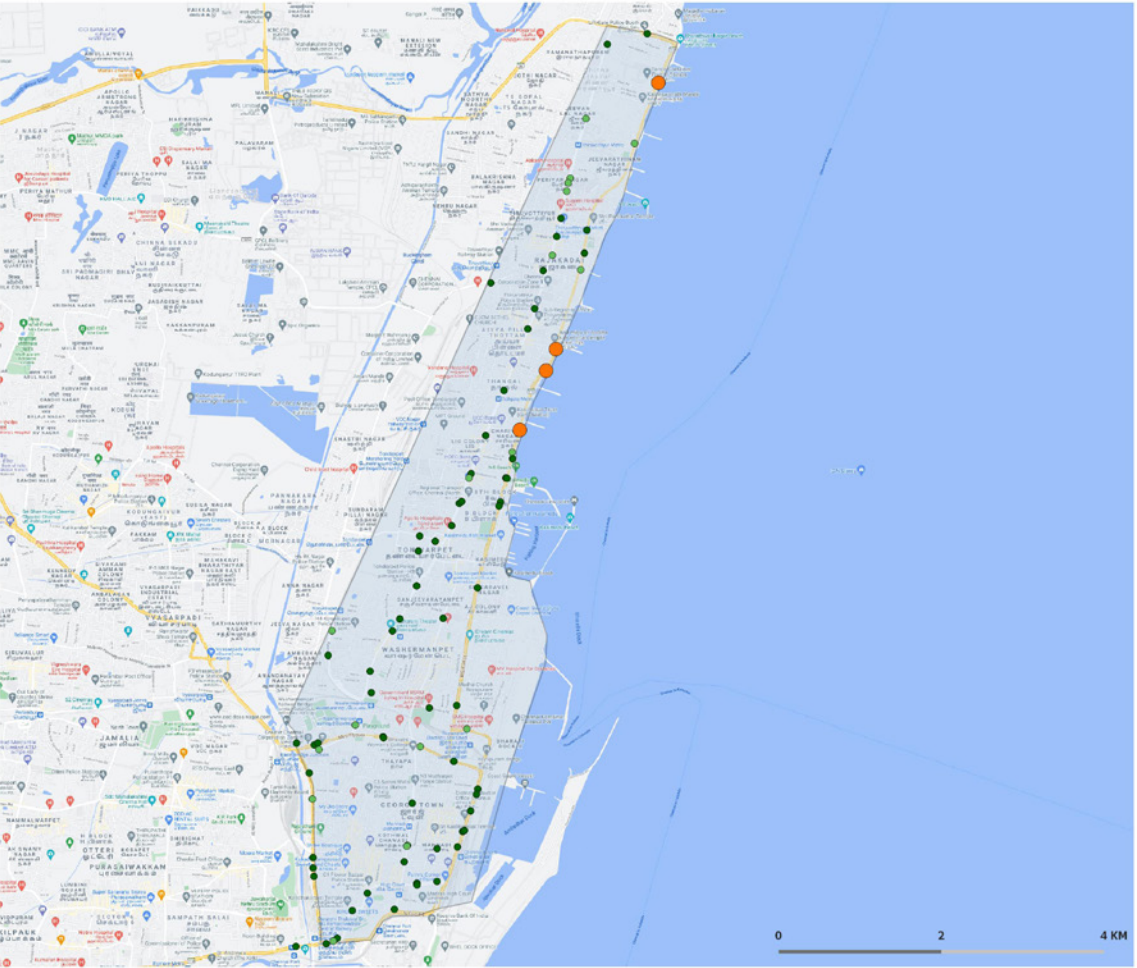
PARAMETER	AUDIT POINTS	RECOMMENDATIONS
	Total audit points with poor rating	Among the four actionable parameters, Lighting is adequate in the wards with 'above average' rating but Walkpath, Visibility & Public Transport need attention to improve overall safety and accessibility in the area.
Walkpath	968	Footpath is inaccessible in most wards because of poor walking condition and unavailability of clear walking space. Removing footpath blockages and having safe walking surfaces will improve accessibility in this area.
Visibility	511	Not many windows and entrances of shops or residences overlook the streets at some locations in these wards. Presence of high boundary walls, isolating the streets is another reason for low visibility. Organising street vendors/informal shopping and street parking will improve visibility in these wards. Lowering opaque boundary walls will also improve visibility.
Public Transport	212	Metro stations, bus stops or auto/rickshaw stands could not be reached within 5-10 minutes of walking distance at few locations in these wards. Adding public transport stands/stops within 2-5 minutes of walking distance will enhance access to public transport and improve overall accessibility in the area.

6. IMPROVE SAFETY AND ACCESS AT BUS STOPS TO ENHANCE USAGE

Data generated by Safetipin's visual mapping was segregated for bus stop locations. The map below shows 92 bus stop locations in Tondiarpet with their safety ratings. 4 locations were found to be below average mainly due to non availability of walkpath and very poor visibility conditions. During the time of mapping the locations were found to be deserted or had very few people present in them. No women were seen in these locations either during mapping. A set of recommendations are given below for better access and safety for these bus stops.

1. Dedicated, unobstructed, paved pedestrian walkpaths must be provided to give people easy and safe access to bus stops (see Walkpath recommendations below).
2. Universal accessibility standards must be met for bus stops to allow people with disability better access to and from the bus stops and while boarding/unboarding the bus.
3. Pedestrian crossings must be appropriately placed near bus stops for safe and easy access to/from stops.
4. Bus stops must be avoided at intersections and should be placed at least 30 meters away from them.
5. Bus stops must have areas allocated near them for first/last mile connectivity transport options. They also must provide cycle stands as a park and ride facility.
6. Whenever possible, bus stops should be located in an area where there is enough visibility either from adjacent buildings and/or from people using that area. Having better visibility enhances the perception of safety among women and other users.

Chennai: Bus Stops Safety Score



7. MAKE LOCAL MARKETS PEDESTRIAN FRIENDLY

Data generated by Safetipin's visual mapping was segregated for local markets. Among five local markets being audited by both applications, three markets were found to have lower safety ratings on Footpaths, Visibility and Public Transport. The map below shows these market locations in Tondiarpet with their safety rating points within 100m and 500m radius. The ratings on the mentioned parameters were poor, below average and average at various audit points mainly due to poor walkpath quality, poor visibility conditions and weak access to public transport. During the time of mapping the locations were found to have very few women and children present in them. Mostly men were seen in these locations during the assessment. A set of recommendations are given below for better access and safety for these markets.

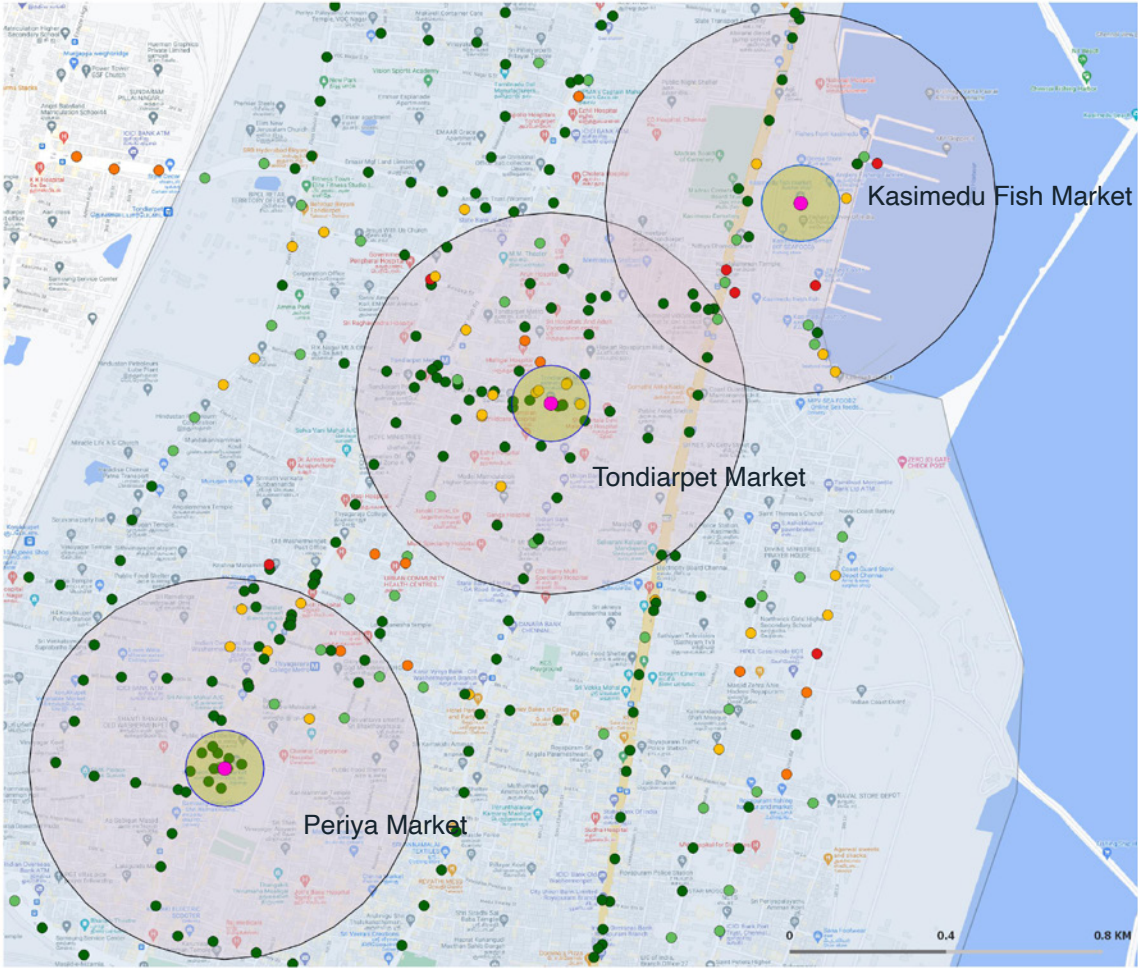
- 1. Dedicated, unobstructed, paved pedestrian walkpaths must be provided to give people easy and safe walking environment within and around local markets (see Walkpath recommendations below).
- 2. Universal accessibility standards must be met for walkpaths to allow people with disability better access within and around local markets (see Walkpath recommendations below).
- 3. Market places should have a parking strategy in place to organise and manage vehicular traffic.
- 4. Markets should have amenities such as public toilets, police assistance booths and resting areas.
- 5. Public transport stops must be with 2-5 mins reach and must have areas allocated near them for first/last mile connectivity transport options.
- 6. Areas around markets must have good visibility either from adjacent buildings and/or from people using that area. Having better visibility enhances the perception of safety among women and other users. Reducing boundary wall heights, front setbacks of plots and encouraging 'active frontage' for better visual connection between the built edge and the pedestrian on the street are some useful strategies to incorporate in order to enhance safety.

The map below show the areas within 500m radius of all the five local markets which were audited by Safetipin applications.

Tondiarpet: Markets Safety Score

Legend

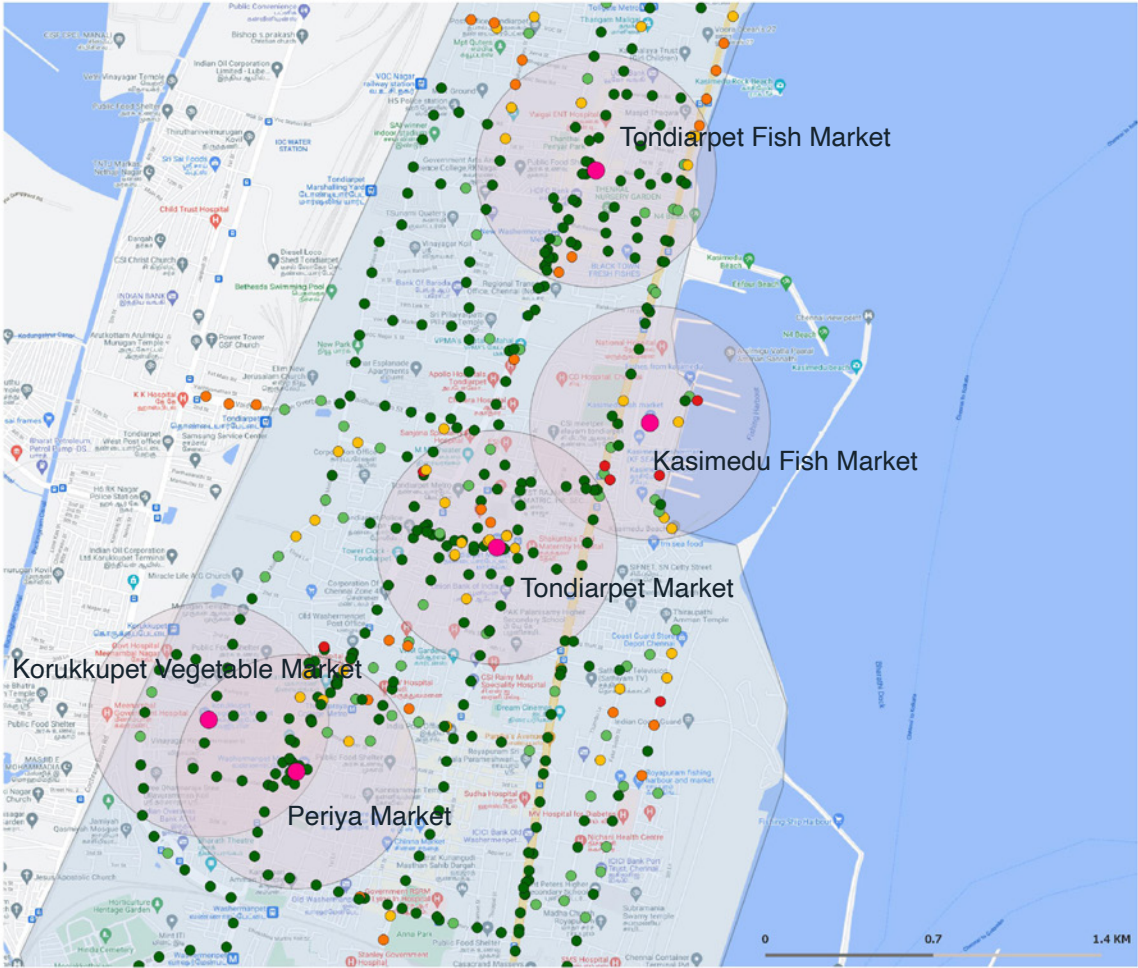
- Market
- Safety Audits
 - Poor
 - Below Average
 - Average
 - Above Average
 - Good
- 100M Radius
- 500M Radius
- Tondiarpet
- Base: Google Map



Tondiarpet: Markets Safety Score

Legend

- Market
- Safety Audits
 - Poor
 - Below Average
 - Average
 - Above Average
 - Good
- 500M Radius
- Tondiarpet
- Base: Google Map



8. IMPROVE SAFETY AND ACCESS AROUND EDUCATIONAL INSTITUTIONS

Data generated by Safetipin's visual mapping was segregated for educational institutions and facilities - schools, colleges and hostels. Among five such places being audited by both applications, two were found to have lower safety ratings on Footpaths, Visibility and Public Transport. The map below shows one school and one college location in Tondiarpet with their safety rating points within 100m and 500m radius. The ratings on the mentioned parameters were poor, below average and average at various audit points mainly due to poor walkpath quality, poor visibility conditions and weak access to public transport. A set of recommendations are given below for better access and safety for these markets.

- 1. Dedicated, unobstructed, paved pedestrian walkpaths must be provided to give people easy and safe walking environment within and around local markets (see Walkpath recommendations below).
- 2. Universal accessibility standards must be met for walkpaths to allow people with disability better access within and around local markets (see Walkpath recommendations below).
- 3. Public transport stops must be with 2-5 mins reach and must have areas allocated near them for first/last mile connectivity transport options. They also must provide cycle stands as a park and ride facility as a inexpensive and sustainable mode of transport for young people.
- 4. Areas around schools and colleges must have good visibility either from adjacent buildings and/or from people using that area. Having better visibility enhances the perception of safety among women and other users. Reducing boundary wall heights, front setbacks of plots and encouraging 'active frontage' for better visual connection between the built edge and the pedestrian on the street are some useful strategies to incorporate in order to enhance safety.

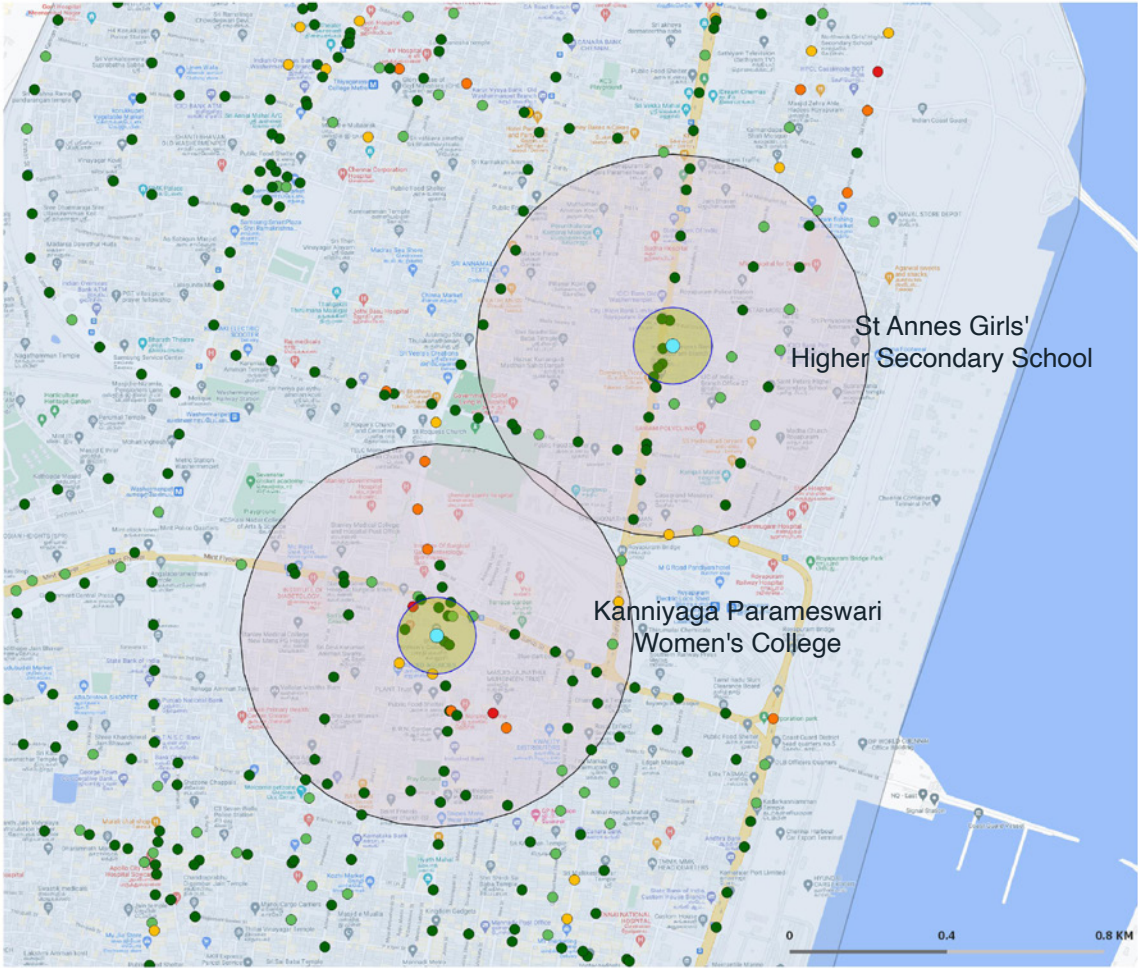
The map below show the areas within 500m radius of all the five educational institutions which were audited by Safetipin applications.

Tondiarpet: School/College Safety Score

Legend

- School/College
- Safety Audits
 - Poor
 - Below Average
 - Average
 - Above Average
 - Good
- 100M Radius
- 500M Radius
- Tondiarpet

Base: Google Map

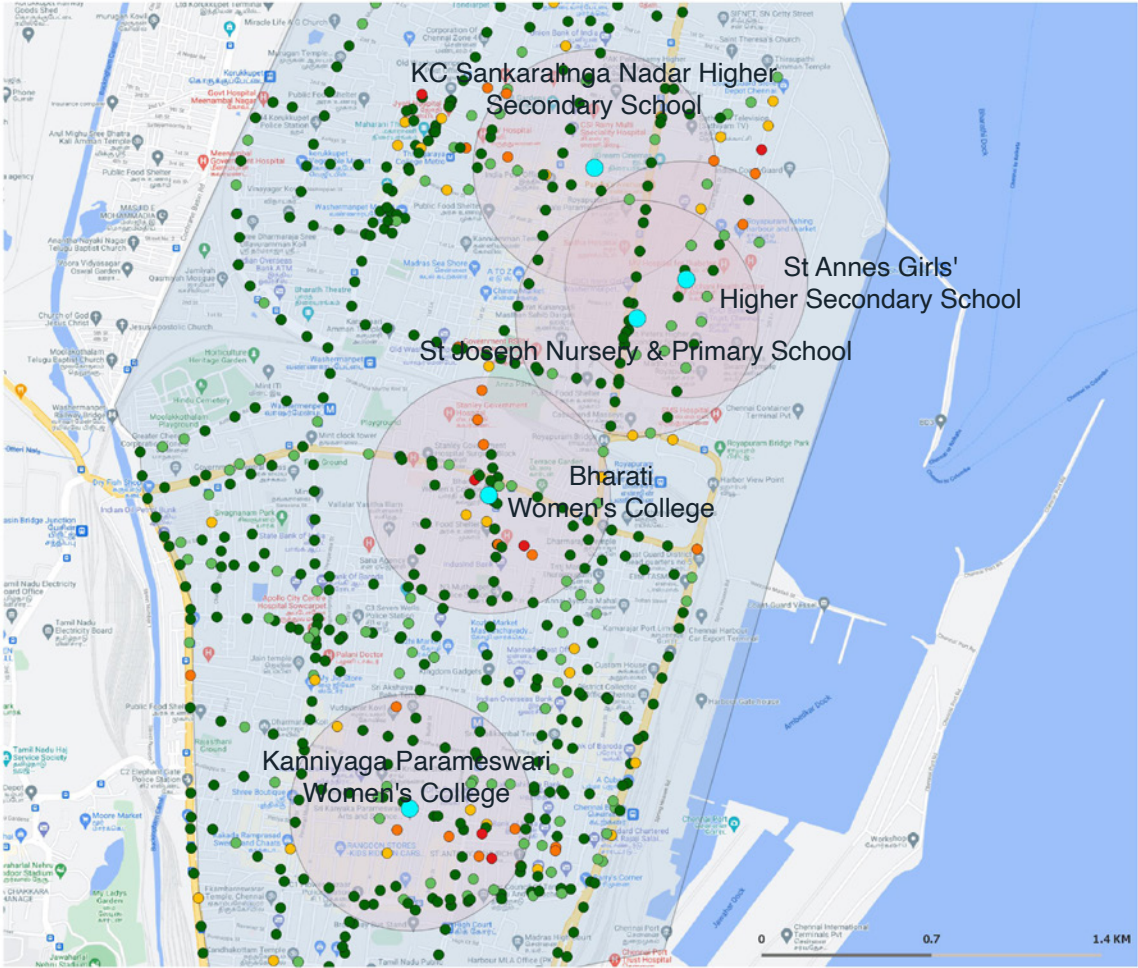


Tondiarpet: School/College Safety Score

Legend

- School/College
- Safety Audits
 - Poor
 - Below Average
 - Average
 - Above Average
 - Good
- 500M Radius
- Tondiarpet

Base: Google Map



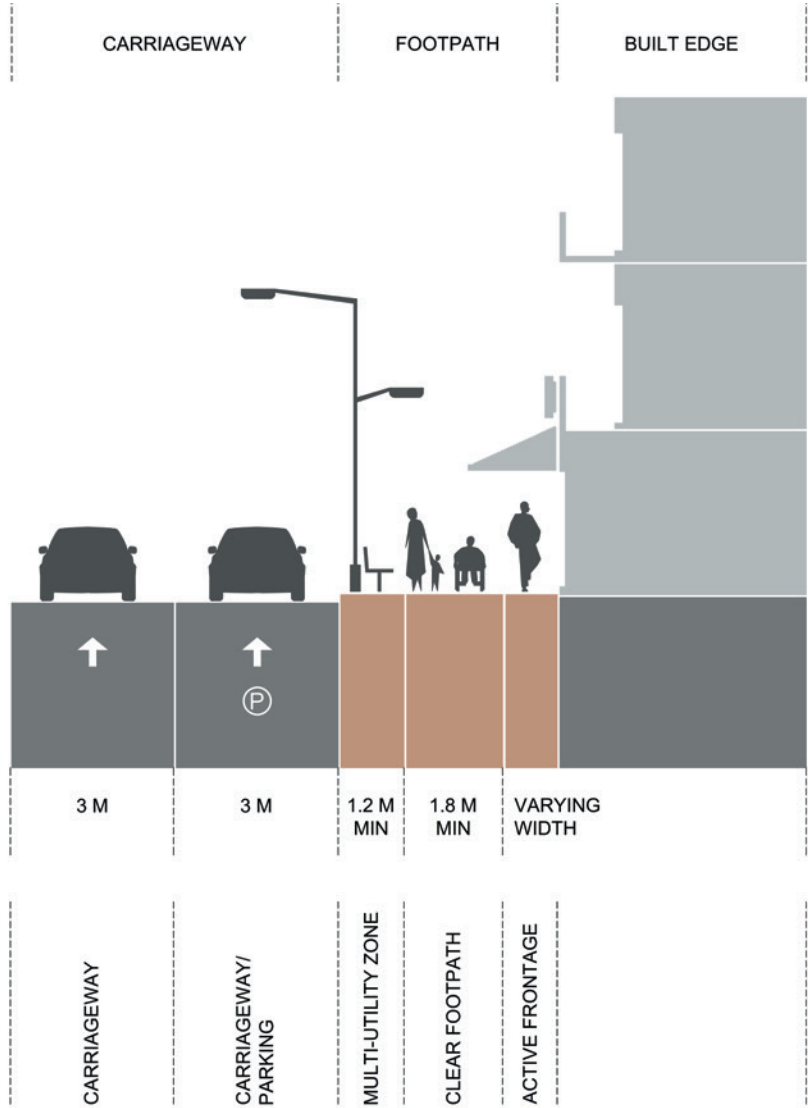
9. IMPROVE WALKABILITY BY ADOPTING INTERVENTION STRATEGIES

Walkpath is the only parameter which fared very poorly on the Safetipin visual analysis. This section includes some strategies which could address the current issues of the walkpath and make it safe, accessible and usable for all.

STRATEGY: REMOVE - REPAIR - RECLAIM

A three tiered action plan of remove, repair and reclaim could be adopted to improve access for all, especially the vulnerable groups - women, children, elderly and people with disability. The table given includes recommendations against each findings to incorporate these strategies. An illustration of a typical street section is given below to explain how the footpaths could be organised for safe access and better utility.

Footpaths have three main components:
Multi-utility Zone – Minimum 1.2 meter footpath width is required as multi-utility space for street utility, vendors and street furniture.
Pedestrian Walkway - Minimum 1.8 meter footpath width is needed for clear pedestrian access - the minimum width required for two people to move or pass each other.
Active Frontage – The area outside shops and houses where spill outs, display windows, entrances, verandas, bay windows, etc. are present. The width could vary depending on space available after allocating space for pedestrians and utilities.



TYPICAL STREET SECTION

Image on right

A typical section of an organised footpath

FINDINGS	RECOMMENDATIONS
REMOVE obstructions out of the footpaths to provide clear walking space	
OBSSTRUCTIONS PRESENT ON THE FOOTPATHS	Images of types of obstructions present on the footpaths can be seen on Page 00
Shop encroachments / house extensions (92)	Remove shop and/or house extensions blocking the existing footpaths to provide clear walking space (min 1.8m). Some spill over from shops could be allowed on the footpaths in the 'active frontage zone' (refer illustration below) depending on available space on the existing footpaths.
Temporary shops / vendors (208)	Remove temporary shops and vendors occupying and blocking the existing footpaths to provide clear walking space (min 1.8m) and organise them on the carriageway side, in the multiutility zone (refer illustration below) part of the footpaths, wherever the footpath width is more than 1.8m - the minimum width required for two people to move. Vendors provide 'eyes on the street' and promote activity on the streets, together which make streets safer. Hence keeping vendors on streets is a mutually beneficial strategy and allocating space for them either on the footpaths or along the footpaths is crucial.
Electricity, signage and signal poles (135)	Remove electricity, signage and signal poles out of the existing footpaths to provide clear walking space (min 1.8m) and arrange/align them on the carriageway side, in the multiutility zone (refer illustration below) part of the footpath, wherever the footpath width is more than 1.8m.
Telephone / communication boxes / power transformers	Remove telephone/communication boxes and power transformers out of the existing footpaths to provide clear walking space (min 1.8m) and arrange/align them on the carriageway side, in the multiutility zone (refer illustration below) part of the footpaths, wherever the footpath width is more than 1.8m.
Garbage dumps and construction debris (93)	Remove garbage dumps and construction debris from the existing footpaths to provide clear walking space (min 1.8m). Plan and organise garbage collection points on the street at various locations to avoid spilling of garbage on the footpaths causing serious hazard and health issues.
Construction material blocking the footpath (167)	Remove construction materials piled on and blocking the existing footpaths to provide clear walking space (min 1.8m). Allocate some space on every street for storing of construction material for all ongoing construction, repair or maintenance work.
Vehicle (two, four wheelers, cycle rickshaws) parked (544)	Remove two & four wheelers, autos, cycles and cycle rickshaws parked on existing footpaths to provide clear walking space (min 1.8m). Organise parking for two, four wheelers, autos, cycles and cycle rickshaws on shared lane (parking and driving) of the carriageway. Refer illustration below.
REPAIR broken footpath and add accessibility elements	
ACCESSIBILITY IMPEDEMENTS OF THE FOOTPATH	Images of types of footpath accessibility elements can be seen on Page 00
Non availability of Curb Ramps	Curb ramps are an essential component of footpaths for improving accessibility for all and could be only found in few locations in Tondiarpet. Add curb ramps to negotiate all level drops and make it step-free to reduce hazards for the vulnerable groups. Refer illustrations on Page 27.
Non availability of Zebra Crossings	Zebra crossings are vital elements of street design to ensure safe pedestrian crossover for all. Many street intersections as well as crossover points in Tondiarpet did not have zebra crossings. Add zebra crossings to all street intersections/crossovers for safe pedestrian access. Refer illustrations on Page 27.
Non availability of Median Breaks	Median breaks facilitate at grade street crossovers for pedestrians and are a crucial component for improving accessibility for all. There was no median break available at the intersections. Add median breaks to facilitate at grade crossover for all vulnerable groups. Refer illustrations on Page 27.
Height of the footpath	Footpath height should be not more than one step (150 mm) high as per international street design standards. Reduce overall footpath height to ensure easy, hazard-free, pedestrian access.
Surface of the footpath	Pedestrian, especially vulnerable groups need at grade walking surfaces. Footpath surfaces with missing, broken, dis-jointed paver blocks, manhole cover protruding out, other fixed objects pose as hazards for pedestrians. Ensure at grade, hazard-free walking surface.
Pedestrain Footover Bridge and Pedestrian Underpass	Pedestrians, especially vulnerable groups need at grade walking surfaces. Footover bridges without lifts are difficult to use by vulnerable groups. Women often carrying children and shopping bags find it difficult to climb up and down such facilities. Pedestrian underpasses have the same issues coupled with being very unsafe after dark. They are often deserted even when they are well lit at night. Investing in high quality, at grade walking surfaces with assessibility elemnets is far more beneficial than building expensive pedestrian footover bridges and underpasses in a dense neighbourhood like Tondiarpet.
RECLAIM vehicular lanes for footpath widening	
INSUFFICIENT SPACE AVAILABLE FOR PEDESTRIANS	Images of insufficient footpath space can be seen on Page 00
Varying footpath widths	Putting pedestrian over vehicle would ensure safe accessibility for all in Tondiarpet. Currently the streets function as shared streets with slower vehicular speed, as people and a range of daily activities occupy the streets. Yet, only able bodied adults are able to navigate these streets and that too with great difficulty. It is even more challenging for vulnerable groups to negotiate the uncertainties encountered frequently on these streets. Hence, providing dedicated footpaths, with clear walking space for pedestrians, on both sides of the streets must be strategized. This could be done by reclaiming parts of the vehicular lanes wherever possible to widen the existing footpaths which do not have adequate clear width for walking. It is important to provide minimum 1.8 meter footpath width for clear pedestrian access - the minimum width required for two people to move or pass each other. Another minimum footpath width of 1.2 meter is needed for multi-utility space to accommodate street utility, vendors and street furniture. Any extra space available after allocating the above two, could be added to 'active frontage' – the space outside the built edge. For illustration, refer typical street section on page 50.
Google Map Link and CSV Files	Please use the map links and the csv files shared in the project to view the collected data supported with images where obstructions, incorrect assessibility elements and insufficient width are present.

URBAN DESIGN STRATEGIES

Planned interventions

Unlike action points which are pragmatic in nature, long term public space intervention strategies look at the city holistically. These strategies serve the purpose of helping cities function better and more effectively. They make the connection between aspects that are linked but not so visible, such as the link between improved physical access and opportunities for a better life for women and others. Below are a few design principles as recommendations which would help Tondiarpet in achieving the overarching goal of 'safe access and participation in public life for women and others.'

1. MAKE WALKING SEAMLESS

Footpaths, curb ramps, crossings, footovers, underpasses

To make walking easy, safe and seamless walking routes must be in good condition and connected to a larger network of streets in the city. A good walking route must be well paved, well lit and free from any type of obstruction. Additionally all walking routes must be well connected to other routes and well designed with crossovers to have a seamless walking experience when travelling from point A to point B. This particularly benefits vulnerable users.

2. MAKE PUBLIC TRANSPORT EASILY ACCESSIBLE

Transport stops, routes, frequency, capacity, safety within transport

To make public transport within everyone's reach, various transportation systems functional in the city should be looked at. It is important to understand how one transport system interacts with the other and how people use them. Only then the gaps or overlaps between systems could be identified and improved to make them seamless. Such interventions to stitch different systems make the overall transport network accessible and allow people to switch within transport systems easily. Safety inside public transport is a very essential aspect to consider for women.

3. FOCUS ON LAST MILE CONNECTIVITY

Non-motorised transport, light motor vehicles, informal transport

Last mile connectivity is an important aspect of any public transportation system operating in the city. Strengthening informal transport like non-motorised vehicles and light motor vehicles which serve as first/last mile connectivity to one's origin and destination could be an useful way of ensuring seamless and safe travelling. Often women forego opportunities available in the city due to weak, unsafe or expensive first/last mile connectivity options.

4. MAKE STREETS ACTIVE

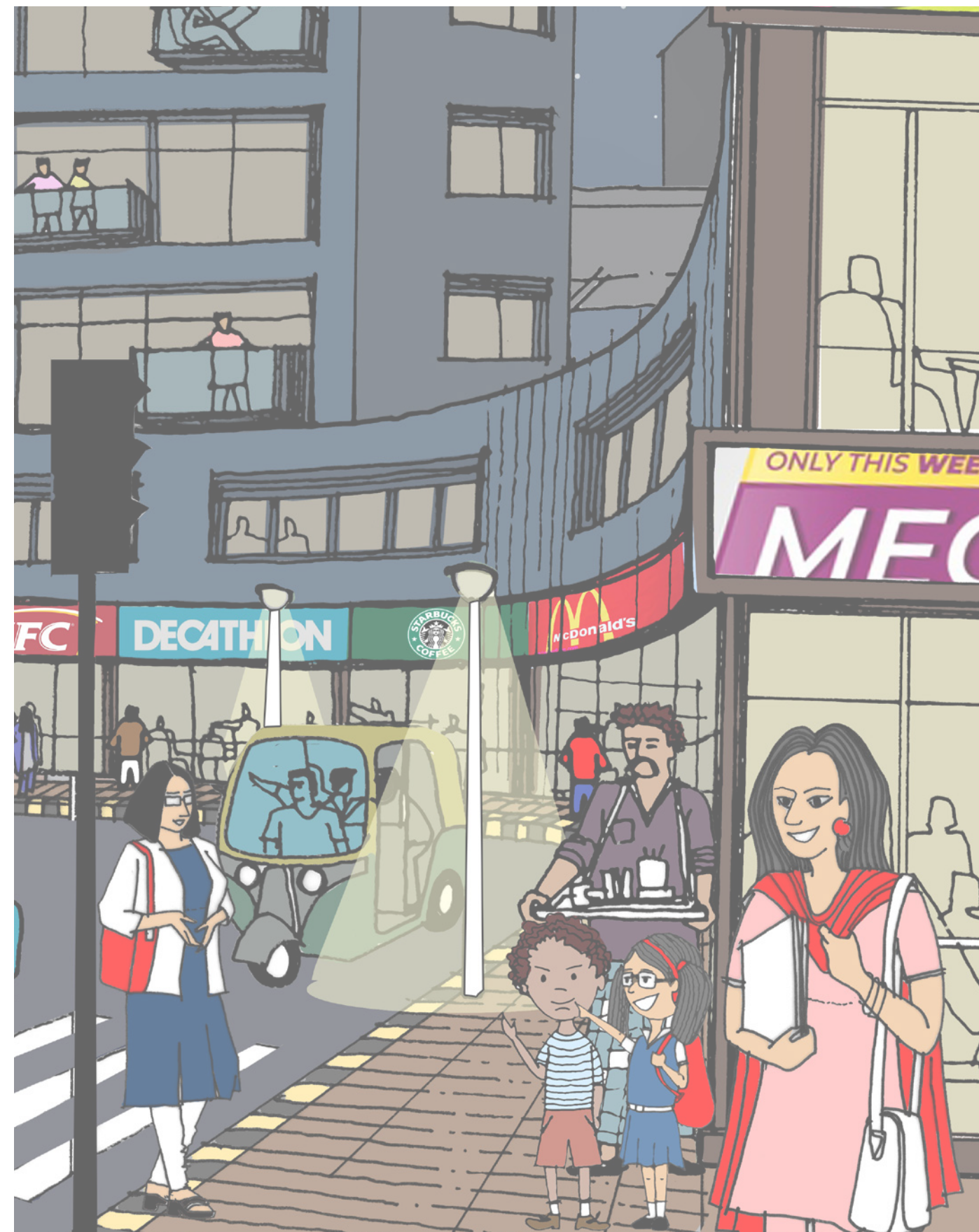
Mixed-use development, informal shopping, eyes on the streets

Active streets are safe streets. To make streets safe an optimum amount of activity on the streets and by the streets is required. Just the way over crowded streets tends to be unsafe, similarly under used streets are unsafe too. Streets which have enough people overlooking it from buildings next to it and have enough people present on it are safer. Encouraging mixed-used developments, on-street shopping, informal hawking and discouraging high, opaque boundary walls and front setback for buildings are good practices to make streets active round the clock.

5. MAKE STREETS INCLUSIVE

Employment in public services, representation in govt, advocacy and sensitisation programs

Ensuring women are employed and represented in public services and government include their perspective in planning, designing and maintaining public space. Participatory planning methods are useful ways of including women and other disadvantaged group's perspective into planning.



APPENDIX

This section includes the CSV files generated for Tondiarpet as examples. These files have geo location for each audit point along with the ratings of every parameter. Every audit point is also accompanied by the images collected at that particular audit location. These files could be integrated with any GIS platform and analysed further. These could be also seen on Google Maps for easier visualisation.

MY SAFETIPIN CSV FILE

S No	ID	Latitude	Longitude	Lighting	Walkpath	Public Transport	Visibility	People	Gender Usage	Security	Openness	Feeling	Safety Score	Comment	Image Link_1	Image Link_2	
1	741868	13.123944	80.2970964	2	0	1	2	1	0	1	2	1	2.6	as this market area sc	http://api-android.s	http://api-android.s	
2	741869	13.123531	80.2973861	2	0	1	2	2	1	0	2	1	3	in one zone after 10C	http://api-android.s	http://api-android.s	
3	741870	13.1241778	80.2968984	2	0	1	2	1	0	2	2	2	4	one zone after 100m	http://api-android.s	http://api-android.s	
4	741871	13.1246292	80.2967513	1	0	1	3	2	0	1	3	1	3.5	only males open can	http://api-android.s	http://api-android.s	
5	741873	13.1255618	80.2969185	1	0	1	1	1	0	0	1	1	0.9	no females present h	http://api-android.s	http://api-android.s	
6	741874	13.124451	80.296936	3	3	3	2	2	1	1	2	2	4.1	police barricade was	http://api-android.s	http://api-android.s	
7	741875	13.1276151	80.297645	2	0	1	3	2	0	0	1	1	2.6	i was standing for few	http://api-android.s	http://api-android.s	
8	741876	13.1285925	80.2980634	3	0	1	2	1	0	0	3	2	4	autos availablemales	http://api-android.s	http://api-android.s	
9	741877	13.1284335	80.2983728	1	0	0	1	1	0	0	0	1	0.7	very few people no fe	http://api-android.s	http://api-android.s	
10	741878	13.1284237	80.2979028	1	0	2	2	1	0	2	2	2	4	lighting pillars there	http://api-android.s	http://api-android.s	
11	741879	13.1284237	80.2979028	2	1	3	2	1	0	0	2	3	4.2	main area entrance c	http://api-android.s	http://api-android.s	
12	741880	13.1180919	80.2931607	3	3	1	2	2	2	2	1	2	4.4	main road so lights a	http://api-android.s	http://api-android.s	
13	741881	13.1179075	80.2931089	2	1	2	3	1	1	1	1	1	2	4.1	pathway available bu	http://api-android.s	http://api-android.s
14	741882	13.1110994	80.2918932	1	2	2	3	2	2	3	3	3	4.6	police station was ve	http://api-android.s	http://api-android.s	
15	741883	13.1104075	80.2919151	2	3	2	2	1	1	2	1	3	4.4	in the entrance gate	http://api-android.s	http://api-android.s	
16	741884	13.1099609	80.2918048	3	3	2	3	1	2	1	2	2	4.4	no stationary shops	http://api-android.s	http://api-android.s	
17	741885	13.1094973	80.2917571	3	3	2	3	2	2	2	1	2	4.5	main road vehicle bu	http://api-android.s	http://api-android.s	
18	741886	13.1095133	80.2917435	3	3	3	3	2	2	2	3	3	4.8	open in all zones veh	http://api-android.s	http://api-android.s	
19	741887	13.1097491	80.2917	3	3	2	2	1	1	2	1	2	4.3	females occupied eq	http://api-android.s	http://api-android.s	
20	741888	13.1080246	80.2915283	3	3	3	3	3	3	3	3	3	5	traffic police availab	http://api-android.s	http://api-android.s	
21	741889	13.1086368	80.2883839	1	0	3	1	2	2	3	1	1	4.1	traffic police availab	http://api-android.s	http://api-android.s	
22	741890	13.1089603	80.2870464	1	0	3	2	2	1	3	3	2	4.3	police booth and ma	http://api-android.s	http://api-android.s	
23	741891	13.1094017	80.2854288	0	0	2	1	1	1	0	1	1	1.3	very less lights peopl	http://api-android.s	http://api-android.s	
24	741892	13.1093163	80.2853883	2	0	2	3	2	1	1	3	2	4.3	moving areas lightin	http://api-android.s	http://api-android.s	
25	741893	13.1257053	80.2878527	0	0	0	0	0	0	0	0	0	0.3	not safetasmac bar w	http://api-android.s	http://api-android.s	
26	741894	13.1257731	80.2879651	3	0	3	1	1	0	0	2	1	3	unsafeno people cro	http://api-android.s	http://api-android.s	
27	741895	13.1235824	80.2878952	3	0	3	3	3	1	3	3	2	4.6	many police are ther	http://api-android.s	http://api-android.s	
28	741896	13.12345	80.2876262	3	0	3	3	2	2	3	3	3	4.7	more people Availab	http://api-android.s	http://api-android.s	
29	741897	13.1233942	80.2879999	2	0	2	2	3	0	2	2	1	4.1	in entrance of the m	http://api-android.s	http://api-android.s	
30	741898	13.1232331	80.2885023	2	0	2	2	2	2	0	1	2	4.1	many shops opned	http://api-android.s	http://api-android.s	
31	741899	13.1228651	80.2887649	3	0	1	2	2	1	0	2	2	4.1	http://api-android.s	http://api-android.s	http://api-android.s	
32	741900	13.1233423	80.2884864	1	0	1	2	2	2	0	2	2	4	females presence wa	http://api-android.s	http://api-android.s	
33	741901	13.1228389	80.2897989	2	0	1	3	2	3	0	3	2	4.3	main Street of the ro	http://api-android.s	http://api-android.s	
34	741902	13.1230698	80.2893569	3	0	1	3	2	2	0	2	2	4.2	all in one place mark	http://api-android.s	http://api-android.s	
35	741903	13.1228596	80.2901906	3	0	1	3	2	1	0	2	2	4.1	space for the road i	http://api-android.s	http://api-android.s	
36	741904	13.1227596	80.2909617	1	0	1	2	1	1	0	2	1	2	visibility is thereope	http://api-android.s	http://api-android.s	
37	741905	13.1227404	80.2909663	2	0	3	3	2	2	0	2	2	4.3	entrance of the mark	http://api-android.s	http://api-android.s	
38	741906	13.1230649	80.2890573	3	0	3	3	2	2	0	2	2	4.3	bus stop very near li	http://api-android.s	http://api-android.s	
39	741907	13.1227681	80.2913782	1	0	2	1	1	1	0	1	2	2.6	very few lightsfemale	http://api-android.s	http://api-android.s	
40	741908	13.1229842	80.2893223	1	0	2	1	1	1	0	1	2	2.6	acceptable lights we	http://api-android.s	http://api-android.s	
41	741909	13.1196154	80.2903986	1	0	3	2	1	1	0	2	2	4	lights was less auto	http://api-android.s	http://api-android.s	
42	741910	13.1195903	80.2903981	3	0	3	3	3	3	3	3	3	4.8	all open place main	http://api-android.s	http://api-android.s	
43	741911	13.1159433	80.2886644	3	0	2	2	1	3	0	1	2	4.1	http://api-android.s	http://api-android.s	http://api-android.s	
44	741912	13.1166174	80.2873133	1	0	2	1	1	1	0	1	1	1.6	females feels comfor	http://api-android.s	http://api-android.s	
45	741913	13.1169595	80.2857347	2	0	1	1	1	1	0	1	1	1.6	narrow streets lights	http://api-android.s	http://api-android.s	
46	741914	13.1169694	80.2855112	3	3	3	2	2	2	0	2	2	4.4	walk path is there be	http://api-android.s	http://api-android.s	
47	741915	13.1175677	80.284509	2	0	3	2	2	0	1	3	4	4.3	females are there in	http://api-android.s	http://api-android.s	
48	741916	13.1175677	80.284509	3	0	3	3	2	2	1	3	3	4.6	park is on closed due	http://api-android.s	http://api-android.s	
49	741917	13.1176958	80.2845659	3	0	3	3	2	2	0	2	3	4.4	not bad things happi	http://api-android.s	http://api-android.s	
50	741918	13.1187599	80.2851749	3	3	3	3	3	1	2	2	3	4.8	side space with lines	http://api-android.s	http://api-android.s	
51	741919	13.1187599	80.2851749	2	0	3	1	2	1	3	1	2	4.2	http://api-android.s	http://api-android.s	http://api-android.s	
52	741920	13.1187195	80.2851255	1	0	3	2	3	1	3	2	3	4.4	police female was th	http://api-android.s	http://api-android.s	
53	741921	13.1185306	80.2850857	2	0	3	2	3	3	2	1	3	4.5	Females were moren	http://api-android.s	http://api-android.s	
54	741922	13.1180821	80.2848001	1	0	3	1	2	0	0	1	2	3	very few lights availa	http://api-android.s	http://api-android.s	
55	741923	13.1175627	80.2845608	3	0	2	1	2	1	2	1	2	4.1	females present in th	http://api-android.s	http://api-android.s	
56	741924	13.116773	80.2839344	3	1	2	2	1	2	1	2	1	2	4.3	some space is there i	http://api-android.s	http://api-android.s
57	741925	13.1161138	80.2835727	2	0	2	3	2	2	1	3	2	4.3	http://api-android.s	http://api-android.s	http://api-android.s	
58	741926	13.1161138	80.2835727	2	0	2	2	2	1	0	2	2	4.1	all 4 places open can	http://api-android.s	http://api-android.s	
59	741940	13.1141252	80.2821959	2	1	2	2	2	2	2	2	2	4.3	female mentioned th	http://api-android.s	http://api-android.s	
60	741941	13.1138863	80.2826295	2	1	2	3	2	2	1	2	2	4.3	many female sellers	http://api-android.s	http://api-android.s	
61	741942	13.11421	80.2828858	1	0	2	2	2	2	1	0	1	2	3.5	more females sellers	http://api-android.s	http://api-android.s
62	741943	13.1145547	80.282843	2	0	3	3	2	2	1	2	2	4.3	many shops open alv	http://api-android.s	http://api-android.s	
63	741944	13.114709	80.2827255	2	0	3	2	2	2	1	2	3	4.4	roads need to be rep	http://api-android.s	http://api-android.s	
64	741946	13.1144308	80.2831058	2	1	3	2	2	2	1	2	3	4.4	road is the major prc	http://api-android.s	http://api-android.s	
65	741947	13.1142906	80.2834085	1	0	2	2	2	2	1	2	2	4.2	no parking for marke	http://api-android.s	http://api-android.s	
66	741948	13.1141924	80.2830834	1	0	2	2	3	0	0	1	2	3.5	only males available	http://api-android.s	http://api-android.s	
67	741949	13.1137284	80.2829251	1	0	2	2	2	2	1	1	2	4.1	if safety camera avail	http://api-android.s	http://api-android.s	
68	741950	13.1141888	80.2827574	2	0	3	2	2	3	2	1	2	4.4	main road adjacent	http://api-android.s	http://api-android.s	
69	741951	13.1154323	80.2827524	2	2	3	2	2	2	1	3	4	4.5	parking space need t	http://api-android.s	http://api-android.s	
70	741952	13.133675	80.2925784	2	1	3	2	2	1	2	3	2	4.4	more space is there	http://api-android.s	http://api-android.s	
71	741953	13.1346297	80.2927127	2	2	2	2	2	2	1	2	2	4.3	safe people always pi	http://api-android.s	http://api-android.s	
72	741954	13.135827	80.2930023	1	2	2	2	1	1	1	1	1	4.1	lights were less parki	http://api-android.s	http://api-android.s	
73	741955	13.1361983	80.2928531	2	1	2	2	2	2	3	1	3	4.4	more females occupi	http://api-android.s	http://api-android.s	
74	741956	13.1342275	80.2926216	1	1	3	2	2	1	2	2	2	4.3	safe lights are less au	http://api-android.s	http://api-android.s	
75	741957	13.1335339	80.2930574	1	0	2	2	1	1	1	1	1	1.6	it's purely seems con	http://api-android.s	http://api-android.s	

SAFETIPIN NITE (DAY) CSV FILE

S No	ID	Latitude	Longitude	Lighting	Walkpath	Public Transport	Visibility	People	Gender Usage	Security	Openness	Feeling	Safety Score	Comment	Image Link_1	Image Link_2
1	740309	13.1579438	80.3043026	3	1	2	2	2	3	0	2	2	4.3	D	https://safetipinima.com/740309	https://safetipinima.com/740309
2	740307	13.1590721	80.3055471	3	1	3	1	2	0	0	3	2	4.2	D	https://safetipinima.com/740307	https://safetipinima.com/740307
3	740173	13.1616995	80.3061801	3	2	3	2	2	0	1	3	2	4.4	D CM	https://safetipinima.com/740173	https://safetipinima.com/740173
4	740171	13.162606	80.3063572	3	2	3	3	2	1	1	2	2	4.4	D ZC	https://safetipinima.com/740171	https://safetipinima.com/740171
5	740169	13.1634233	80.3064644	3	2	3	2	2	1	0	2	2	4.2	D	https://safetipinima.com/740169	https://safetipinima.com/740169
6	740167	13.1643113	80.3066629	3	2	3	2	2	3	0	2	2	4.4	D GB CD	https://safetipinima.com/740167	https://safetipinima.com/740167
7	740165	13.1651209	80.3069741	3	2	2	3	2	2	0	2	2	4.4	D	https://safetipinima.com/740165	https://safetipinima.com/740165
8	740163	13.1657895	80.3073925	3	2	3	2	1	0	0	2	2	4.2	D	https://safetipinima.com/740163	https://safetipinima.com/740163
9	740161	13.1664332	80.3079772	3	2	3	2	2	1	0	2	2	4.3	D	https://safetipinima.com/740161	https://safetipinima.com/740161
10	740251	13.1666357	80.3083152	3	1	3	2	0	0	0	2	1	3.5	D	https://safetipinima.com/740251	https://safetipinima.com/740251
11	740159	13.1670536	80.3085834	3	1	3	2	2	3	0	2	2	4.4	D	https://safetipinima.com/740159	https://safetipinima.com/740159
12	740157	13.1677587	80.309608	3	2	3	1	1	0	1	2	2	4.2	D	https://safetipinima.com/740157	https://safetipinima.com/740157
13	740155	13.1683437	80.3101498	3	2	3	1	0	0	0	2	1	3.5	D	https://safetipinima.com/740155	https://safetipinima.com/740155
14	740153	13.1689966	80.3105897	3	2	2	2	0	1	0	2	1	3	D GB	https://safetipinima.com/740153	https://safetipinima.com/740153
15	740151	13.1698637	80.3109598	3	2	2	2	0	0	0	2	1	3.5	D	https://safetipinima.com/740151	https://safetipinima.com/740151
16	740149	13.1707256	80.3112549	3	2	3	0	0	0	0	2	1	3	D	https://safetipinima.com/740149	https://safetipinima.com/740149
17	740147	13.1716344	80.3115284	3	2	3	0	0	0	0	2	1	3	D GB	https://safetipinima.com/740147	https://safetipinima.com/740147
18	740145	13.1725224	80.3118181	3	2	3	0	2	0	0	2	1	4	D	https://safetipinima.com/740145	https://safetipinima.com/740145
19	740110	13.1735723	80.3122419	3	2	3	0	1	0	0	2	1	3.5	D	https://safetipinima.com/740110	https://safetipinima.com/740110
20	740111	13.1745438	80.3125638	3	1	2	0	0	0	0	2	1	2	D	https://safetipinima.com/740111	https://safetipinima.com/740111
21	740112	13.1756563	80.3129715	3	2	3	1	1	0	0	2	1	4	D	https://safetipinima.com/740112	https://safetipinima.com/740112
22	740102	13.176748	80.3135186	3	1	3	0	1	0	0	2	1	3	D	https://safetipinima.com/740102	https://safetipinima.com/740102
23	740100	13.1777613	80.3138673	3	2	3	2	2	0	0	2	2	4.3	D	https://safetipinima.com/740100	https://safetipinima.com/740100
24	740098	13.1787014	80.3141356	3	2	3	0	0	0	0	2	1	3.5	D	https://safetipinima.com/740098	https://safetipinima.com/740098
25	740096	13.1795789	80.3145164	3	2	3	0	0	0	0	2	0	2	D	https://safetipinima.com/740096	https://safetipinima.com/740096
26	740094	13.1804773	80.3149402	3	2	3	0	1	0	0	2	1	3.5	D	https://safetipinima.com/740094	https://safetipinima.com/740094
27	740088	13.1830261	80.3161955	3	1	3	1	2	0	3	3	2	4.4	D	https://safetipinima.com/740088	https://safetipinima.com/740088
28	740113	13.1838827	80.3159326	3	1	3	1	1	0	1	2	1	4	D CD	https://safetipinima.com/740113	https://safetipinima.com/740113
29	740114	13.1842222	80.3149724	3	1	3	1	2	0	2	2	2	4.3	D	https://safetipinima.com/740114	https://safetipinima.com/740114
30	740115	13.1843946	80.3145379	3	1	3	0	2	0	0	2	2	4.2	D	https://safetipinima.com/740115	https://safetipinima.com/740115
31	740116	13.1846296	80.3136796	3	1	3	1	2	0	3	2	2	4.3	D ZC	https://safetipinima.com/740116	https://safetipinima.com/740116
32	740136	13.1848124	80.3130359	3	1	3	2	1	3	3	2	2	4.5	D ZC	https://safetipinima.com/740136	https://safetipinima.com/740136
33	740135	13.1850213	80.3120166	3	3	2	0	0	0	2	2	1	4	D	https://safetipinima.com/740135	https://safetipinima.com/740135
34	740134	13.1852146	80.3110886	3	3	2	0	0	0	1	2	1	3	D	https://safetipinima.com/740134	https://safetipinima.com/740134
35	740133	13.1854444	80.3100318	3	3	2	0	0	0	0	1	2	2	D	https://safetipinima.com/740133	https://safetipinima.com/740133
36	740132	13.1856638	80.3090823	3	3	0	0	0	0	0	2	1	2	D	https://safetipinima.com/740132	https://safetipinima.com/740132
37	740131	13.1858257	80.3081703	3	3	0	0	0	0	0	2	1	2	D	https://safetipinima.com/740131	https://safetipinima.com/740131
38	740130	13.1860555	80.3070813	3	3	0	0	0	0	0	2	1	2	D	https://safetipinima.com/740130	https://safetipinima.com/740130
39	740129	13.1862487	80.306164	3	3	0	0	0	0	0	2	1	2	D	https://safetipinima.com/740129	https://safetipinima.com/740129
40	740128	13.1863793	80.3051662	3	3	0	0	0	0	0	2	1	2	D	https://safetipinima.com/740128	https://safetipinima.com/740128
41	740127	13.1862331	80.3042275	3	2	0	0	0	0	0	2	1	1.6	D	https://safetipinima.com/740127	https://safetipinima.com/740127
42	740126	13.1861391	80.3036803	3	3	0	0	0	0	0	2	1	2	D	https://safetipinima.com/740126	https://safetipinima.com/740126
43	740125	13.186489	80.304544	3	3	0	0	0	0	0	2	1	2	D	https://safetipinima.com/740125	https://safetipinima.com/740125
44	740124	13.1864681	80.3035686	3	3	0	0	0	0	0	2	1	2	D	https://safetipinima.com/740124	https://safetipinima.com/740124
45	740123	13.1864958	80.3064666	3	3	0	0	0	0	0	2	1	2	D	https://safetipinima.com/740123	https://safetipinima.com/740123
46	740122	13.1860764	80.307489	3	3	0	0	1	3	0	2	2	4.1	D	https://safetipinima.com/740122	https://safetipinima.com/740122
47	740121	13.1858988	80.3084654	3	3	0	0	1	3	0	2	2	4.1	D	https://safetipinima.com/740121	https://safetipinima.com/740121
48	740120	13.1856586	80.3093451	3	3	0	0	0	0	0	2	1	2	D	https://safetipinima.com/740120	https://safetipinima.com/740120
49	740119	13.1854183	80.310595	3	3	0	0	0	0	0	2	1	2	D	https://safetipinima.com/740119	https://safetipinima.com/740119
50	740118	13.1852198	80.3115606	3	3	3	2	2	1	2	2	2	4.5	D	https://safetipinima.com/740118	https://safetipinima.com/740118

